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Deep Local Conformal Autoencoders for Semi-Supervised Acoustic Scene Mapping and Speaker Tracking

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Contents

Abstract i

1. Introduction 1
1.1. Research Motivation . . . 1
1.2. Goals and Existing Solutions . . . 1
1.2.1 Acoustic Scene Mapping . . . 2
1.2.2 Dynamic Speaker Tracking . . . 2

I. Two-Dimensional Acoustic Scene Mapping 4

2. Background 5
2.1. Problem Statement . . . 5
2.2. Acoustic Features . . . 5
2.3. LOCA . . . 6

3. Proposed Algorithm - Acoustic Scene Mapping 9
3.1. Data Collection . . . 9
3.2. Training Framework . . . 9
3.3. Inference Framework . . . 11

4. Experimental Study 11
4.1. Acoustic Scene Mapping in a Simulated Room . . . 11
4.2. Out-of-Sample Extrapolation Capabilities . . . 17

II. One-Dimensional Tracking 19

5. Problem Formulation 20
5.1. Problem Statement . . . 20
5.2. Instantaneous Acoustic Features . . . 20

6. Proposed Approach 21
6.1. Modified LOCA . . . 21
6.2. Synthetic Aperture Paradigm for Bursts Collection . . . 22
6.3. Model Training and Parameters . . . 25

7. Experimental Study 26
7.1. Evaluation and Performance Measures . . . 26
7.2. Simulation I - Method Comparison . . . 27
7.2.1 Training Data . . . 27
7.2.2 Test Data . . . 28

7.2.3 Competing Methods 29
7.2.4 Results and Discussion 29
7.3. Simulation II - Number of Anchored Speakers 29
7.4. Simulation III - Generalization to Test Context-Frames 31
7.5. Simulation IV Multi-Scenario Generalization 32
7.5.1 Training Data 33
7.5.2 Test Data 34
7.5.3 Results and Discussion 34
7.6. Ablation Study 34

8. Conclusions **37**
8.1. Summary 37
8.2. Future Directions 37

References **39**

Hebrew Abstract **א**

List of Figures

2.1 Problem Statement 5
2.2 An illustration of LOCA adapted to acoustic scene mapping 8
3.1 A visualization of the problem setup and the training scheme used for acoustic scene mapping 10
4.1 Room sampling strategy visualization 12
4.2 A visualization of the RTF input features 13
4.3 2-D geometric reconstruction of the training net grid achieved by LOCA . . 15
4.4 A comparison between the embeddings achieved for the training net grid by all methods 16
4.5 A demonstration of the isometry measure of the different methods at the case of $\mathrm{RT}_{60} = 160$ ms 17
4.6 A visualization of the extrapolation capabilities of LOCA and A-DM for $\mathrm{RT}_{60} = 160$ ms 18
5.1 An illustration of the experimental setup 20
6.1 An illustration of the modified LOCA framework training scheme 23
6.2 An illustration of the synthetic aperture paradigm used for collecting bursts of measurements 27
7.1 A comparison between the RMSE and accuracy measures for the different acoustic settings 30
7.2 Performance comparison of the modified LOCA model that was trained on speech data (*LOCA-Speech*) when evaluated over various amounts of anchored speakers 31
7.3 Performance comparison of the modified LOCA model that was trained on speech data (*LOCA-Speech*) when evaluated over different amounts of test context frames (N_c^{test}) 32
7.4 An abstract visualization for Simulation (7.5) 33
7.5 An ablation study on the stability of training for the different training approaches of LOCA 36

List of Tables

4.1 MAE (cm) of reconstruction using test data. 15
7.1 Comparison between modified local conformal autoencoder (LOCA) and isometrically regularized variational auto encoder (IRVAE) in the multi-scenario generalization experiment 35

List of Abbreviations

A-DM anisotropic diffusion maps

CCA Canonical Correlation Analysis

CNN convolutional neural network

CRNN convolutional recurrent neural network

DM diffusion maps

DNN deep neural network

DOA direction of arrival

DR deep ranking

EKF extended Kalman filter

FCN fully connected network

GCC-PHAT generalized cross-correlation phase transform

HMM hidden Markov model

iRTF instantaneous relative transfer function

IRVAE isometrically regularized variational auto encoder

LOCA local conformal autoencoder

LSTM long short-term memory

MAE mean absolute error

MMKF multiple model Kalman filter

MRL manifold regularization learning

PHD probability hypothesis density

RADAR radio detection and ranging

RIR room impulse response

RMSE root mean square error

RoI region of interest

RTF relative transfer function

SLAM simultaneous localization and mapping

SNR signal to noise ratio

SRP-PHAT steered-response power phase transform

TDOA time difference of arrival

UKF unscented Kalman filter

VAD voice activity detector

VAE variational auto encoder

WGN white Gaussian noise

WKF wrapped Kalman filter

List of Publications

Conference Publications

[1] Idan Cohen, Sharon Gannot, and Ofir Lindenbaum. "Unsupervised Self-Localization and Mapping based on Acoustic Measurements and Dimensionality Reduction". *In: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2024, pp. 386–390

Abstract

From smartphones and smart speakers to hearing aids and surveillance systems, microphone-equipped devices are, nowadays, everywhere. As a result, there is an increasing demand for advanced audio signal processing techniques to interpret acoustic information in diverse environments accurately. One of the eminent challenges in this field is the presence of reverberation, which can severely distort audio signals and degrade the performance of traditional acoustic processing methods. Accurate acoustic scene mapping and dynamic speaker tracking in such reverberant environments are crucial for applications ranging from augmented reality to robotics and assistive technologies.

The first part of this thesis addresses acoustic scene mapping, aiming to map environments using only the acoustic signals an observer receives. Classical methods for acoustic scene mapping and dynamic speaker tracking often rely on time difference of arrival (TDOA) estimation, which is highly sensitive to reverberation and additive noise. In static acoustic settings, we assert that variations in acoustic features stem mainly from the positional changes of the receiver, hinting that these high-dimensional acoustic features lie on a low-dimensional manifold. Capitalizing on this idea, we apply a novel data-driven, unsupervised manifold learning tool. We propose using LOCA, a deep neural network-based scheme for manifold learning, to exploit the natural structure of the data. Our unsupervised approach uses a microphone array to measure transmitted sound sources at various locations within an acoustic enclosure. We demonstrate that our method learns an isometric representation of the receiver's spatial locations, allowing for extrapolation to new, unvisited regions. The robustness of this approach is validated through realistic simulations, showing significant resilience to reverberation compared to classical and other dimensionality-reduction methods.

The second part of this thesis addresses dynamic speaker tracking, where we aim to track the speaker's direction of arrival (DOA) in real time in noisy and reverberant environments. We present a semi-supervised modified version of the LOCA model together with a novel synthetic aperture training paradigm. Using this methodology, we leverage the strengths of LOCA by utilizing the dynamic nature of the problem. Extensive simulations confirm the effectiveness of our method in dynamic settings with varying levels of noise and reverberation while also showcasing its generalization ability to new settings.

1 Introduction

1.1 Research Motivation

Many modern applications, such as augmented reality and robot autonomy, require a better understanding of the geometry of our surroundings and our position within them. With the increasing availability of cameras and optical sensors, most research has focused on computer vision algorithms for visual mapping and localization [2]. However, as audio devices like mobile phones and hands-free communication systems have become more widespread, the audio processing community has recognized that essential spatial information can also be extracted from audio signals [3, 4].

The main challenge that classical algorithms face in acoustic mapping or localization is the complex propagation of sound waves in reverberant enclosures. In such environments, sound wave propagation doesn't include only the direct path wave but also numerous reflections originating from surfaces and objects within the enclosure. Classical algorithms aim to find the direct path and require the estimation of the TDOA between microphones while ignoring the reflection [5].

In contrast, our study focused on a data-driven approach that uses the entire reflection pattern of signals to extract standardized data coordinates from measurements. Our approach utilizes the LOCA algorithm, a dimensionality reduction scheme for learning a low-dimensional manifold from its high-dimensional representation. We will leverage the natural structure of the acoustic features, specifically the relative transfer functions (RTFs), which are points on a low-dimensional manifold represented in high-dimensional space [6]. It is worth noting that obtaining labeled data in the real world is often a challenge as it requires significant time and effort. Therefore, we will primarily use semi/weakly-supervised algorithms, requiring only a small set of measurements with calibrated positions.

1.2 Goals and Existing Solutions

In our research, we will focus on two different objectives. The first part of our thesis will deal with *acoustic scene mapping*, aiming to recover the region of interest (RoI) within an acoustic enclosure where a microphone-array mounted device traveled in the presence of active speakers. We rely on the fact that the device's position in our static acoustic environment is the only parameter that induces variability over our acoustic features. The second part of our thesis will deal with *dynamic speaker tracking*, aiming to find the 1-dimensional direction of arrival (DOA) w.r.t. a fixed microphone array. Here, the roles are reversed, as the microphone array is fixed, but the speaker is free to move. This time, we suggest that the only parameter that induces variability over our acoustic features is the position of the

speaker.

1.2.1 Acoustic Scene Mapping

Nowadays, a lot of applications require the ability to map our surroundings without any prior knowledge. This process is known as scene mapping, which is closely related to simultaneous localization and mapping (SLAM). In SLAM, the objective is to reconstruct both the trajectory and the environment's map while an observer moves along an unknown path. Acoustic scene mapping, on the other hand, uses audio signals acquired by a microphone array to map the environment.

The traditional way of specifying an environment map is by using landmarks, and reconstructing the map involves locating these landmarks [2]. In our case, we emit audio signals from sound sources whose positions are assumed to be unknown to us and record them using a microphone array that scans the RoI and maps it.

Acoustic SLAM methods require the use of localization techniques, usually based on TDOA estimation between pairs of microphones [3, 4]. The most commonly used approaches for TDOA estimation are the generalized cross-correlation phase transform (GCC-PHAT) algorithm [7] and its multichannel extension, the steered response power phase-transform (SRP-PHAT) [8]. However, these algorithms perform poorly in highly reverberant environments [5], resulting in inaccurate TDOA estimates and poor source localization, particularly for large source-sensor distances. In [4], for example, the authors use a probabilistic approach for localizing both the sound sources and the receivers to account for the erroneous DOA estimates.

Although several improvements to generalized cross-correlation schemes have been proposed [9, 10], simple TDOA-based mappings still cannot reliably reconstruct the environment map.

1.2.2 Dynamic Speaker Tracking

Speaker DOA estimation and, in particular, speaker tracking are essential tasks in numerous domains, ranging from automated conferencing solutions [11, 12] to advanced surveillance systems [13, 14]. With precise speaker tracking, these technologies can enhance sound capture quality, facilitate more efficient communication, and improve security measures.

Classical solutions for this problem, namely the GCC-PHAT [7] and its multichannel extension, the steered-response power phase transform (SRP-PHAT) [15], rely on analyzing the cross correlation between the signals and extracting their TDOA. Other methods, such as the multiple signal classification (MUSIC) algorithm [16] and its extensions, offer high-resolution capabilities and rely on the analysis of the covariance matrix of the received signals. These techniques, which were initially designated to address single-path static scenarios, demonstrate limitations when dealing with the dynamic multi-path case. In [17] and

[18], a grid search was conducted over the output power of a beamformer steered to all potential locations in order to estimate the position. Several recursive least squares-based tracking methods were presented in [10], and probabilistic approaches were presented in [19, 20, 4].

In a prevalent probabilistic approach for speaker tracking, it was suggested to model the source path as a stochastic process, enabling the utilization of Bayesian inference frameworks. In this approach, the TDOAs of the sources are estimated, resulting in nonlinear state-space model equations. Multiple algorithms were proposed in this context, including the Kalman filter [21] and its nonlinear extensions, specifically the extended Kalman filter (EKF) [22, 23], the unscented Kalman filter (UKF) [22], the multiple model Kalman filter (MMKF) [24], and the wrapped Kalman filter (WKF) [25]. Particle filters were also suggested as a viable solution [26, 27, 28, 29], together with hidden Markov models (HMMs) [30, 31], probability hypothesis density (PHD) filters [32, 33, 34, 35] and factor graphs [36]. Similarly, non-Bayesian approaches, based on the recursive expectation maximization (EM) procedure, were suggested at [37, 38, 39, 40, 41]. Recently, greater focus has been directed towards training-based localization methods. These approaches are usually divided into two groups - supervised and semi/weakly-supervised. In the supervised schemes, our training dataset consists of samples whose positions are accurately known. While achieving excellent performance, the supervised approaches require extensive amounts of labeled data, which is often a laborious task to collect. Such approaches include [42] and [43], which suggest using convolutional neural networks (CNNs) and fully connected networks (FCNs) over the acoustic features, respectively. In [44], the authors suggest improving the former approach by integrating a long short-term memory (LSTM) network to exploit the temporal correlation between consecutive time frames. Other works, such as [45] and [46] suggest using a convolutional recurrent neural network (CRNN). In this case, the recurrent layers model the evolution of spatial parameters given the sequential features and their spatial trajectory. In [47], the authors suggest applying a supervised U-Net structure with a dense block to denoise the GCC-PHAT components before constructing the SRP-PHAT, achieving improved tracking.

In the second group, which includes semi/weakly supervised approaches, a deeper understanding of the data's nature is required to exploit the labeled information efficiently. In [48] and [49], a graph-Laplacian-based manifold learning approach is used to learn the latent manifold that controls the acoustic data. In [50], a variational auto encoder (VAE) is used to learn the probability distribution that characterizes the acoustic data. In [51], a weakly-supervised setting is considered for localization, where additional information regarding the relative distance of samples is given in the form of triplets. The triplets consist of an anchor, a positive, and a negative sample - based on their proximity to the anchor. This work was later extended to the dynamic case in [52].

Part I
Two-Dimensional Acoustic Scene Mapping

2 Background

2.1 Problem Statement

We assume we have an acoustic enclosure with a single or multiple active sound sources, fixed in unknown positions. We also assume that the sound sources are not concurrently active, meaning that we can extract the acoustic features induced by each one of them - individually. A device which is equipped with microphone array is free to travel inside the enclosure and record the sources. Our task is then to understand the nature of the enclosure using merely the audio recordings. Explicitly, we wish to reconstruct the area where the device was traveling.



Figure 2.1: Problem Statement: A device mounted with a special microphone array travels in an acoustic enclosure with active sound sources.

2.2 Acoustic Features

The RTFs [53, 54] are used as the acoustic feature for our approach. They are independent of the source signal and carry relevant spatial information. Consequently, they can serve as *spatial fingerprints* that characterize the positions of each of the sources in a reverberant enclosure [55].

RTF - Definition and Estimation: For a pair of microphones, we consider time-domain acoustic recordings of the form

$$x_i(n) = \{a_i * s\}(n) + u_i(n), \tag{1}$$

with $s(n)$ being the source signal, $a_i(n)$, the room impulse responses (RIRs) relating the source and each of the microphones, $i = \{1,2\}$, and $u_i(n)$ noise signals independent of the

source. We define the acoustic transfer functions $A_i(k)$ as the Fourier transform of the RIRs $a_i(n)$. Following [53], the RTF is defined as

$$H(k) = \frac{A_2(k)}{A_1(k)} \tag{2}$$

Assuming negligible noise and selecting $x_1(n)$ as the reference signal, the RTF can be estimated from:

$$\hat{H}(k) = \frac{\hat{S}_{x_2x_1}(k)}{\hat{S}_{x_1x_1}(k)} \approx H(k), \tag{3}$$

where $\hat{S}_{x_1x_1}(k)$ and $\hat{S}_{x_2x_1}(k)$ are the power spectral density (PSD) and the cross-PSD, respectively.

Representing RTFs on the Acoustic Manifold: Previous works have studied the nature of the RTFs [6, 55, 56]. RTFs are high-dimensional representations in $\mathbb{C}^D$ that correspond to the vast amount of reflections from different surfaces characterizing the enclosure. Following the manifold hypothesis [57], we assume that the RTF vectors, drawn from a specific RoI in the enclosure, are not spread uniformly in the entire space of $\mathbb{C}^D$. Instead, they are confined to a compact nonlinear manifold $\mathscr{M}$ of dimension d, much smaller than the ambient space dimension, i.e., $d \ll D$. This assumption is justified by the fact that perturbations of the RTF are only influenced by a small set of parameters related to the physical characteristics of the environment. These parameters include the enclosure dimensions, its shape, the surfaces' materials, and the positions of the microphones and the source. Moreover, we focus on a static configuration, where the enclosure properties and the source position remain fixed. In such an acoustic environment, the only varying degree of freedom is the location of the microphone array.

Although nonlinear, in small neighborhoods, the manifold *is* locally linear, meaning that it is flat in the vicinity of each point and coincides with the tangent plane to the manifold at that point. Hence, the Euclidean distance can faithfully measure affinities between points that reside close to each other on the manifold. For remote points, however, the Euclidean distance is meaningless. This property is utilized to our advantage by LOCA, which makes use of the linear relation between points in a local neighborhood. A visual inspection of the RTFs in our setting is shown later in Fig. 4.2.

2.3 LOCA

Motivation: Our approach requires a localized sampling strategy denoted *burst sampling* [58]. A burst is a collection comprising samples taken from a local neighborhood. Consequently, bursts can provide information on the *local* variability in the vicinity of each data point, allowing for estimating the Jacobian (up to an orthogonal transformation) of the unknown measurement function.

Assumptions and Derivation: First, for simplicity, we consider the case where the latent domain of our system of interest is a path-connected domain in the Euclidean space $\mathscr{X} \subset \mathbb{R}^d$. Observations of the system consist of samples captured by a measurement device and are given as a nonlinear smooth and bijective function $f : \mathscr{X} \to \mathscr{Y}$, where $\mathscr{Y} \subset \mathbb{R}^D$ is the ambient or measurement space, and $D \gg d$. The high dimensional acoustic features, namely the RTFs, constitute the measurement space in our setting. Consider N RTF data points, denoted $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in \mathbb{R}^d$ in the latent space. Assume that all these points lie on a path-connected, d-dimensional subdomain of $\mathscr{X}$. Importantly, we do not have direct access to the latent space $\mathscr{X}$. Samples in the latent space $\mathscr{X}$, which can be thought of as latent states, are pushed forward to the ambient space $\mathscr{Y}$ via the unknown deformation f. Let $\{\boldsymbol{y}_i\}_{i=1}^N \in \mathbb{R}^D$ be a set of measurements, with $\boldsymbol{y}_i = f(\boldsymbol{x}_i)$. We assume that an observed burst around each $\boldsymbol{y}_i$ consists of perturbed versions of the latent state $\boldsymbol{x}_i$, pushed through the unknown deformation f. Formally, for fixed $1 \leq i \leq N$, we define the burst (sample cloud) $\boldsymbol{Y}_i = \{\boldsymbol{y}_i^{(j)}\}_{j=1}^M$ as the set of independent and identically distributed samples of the random variable $f(\boldsymbol{X}_i) \in \mathbb{R}^D$, where $\boldsymbol{X}_i \sim \mathcal{N}(\boldsymbol{x}_i, \sigma^2 \boldsymbol{I}_d)$, are independent random variables and M denotes the number of samples in each burst. The available data now consists of N bursts of observed states and their perturbed samples. We assume that σ is sufficiently small such that, practically, the differential of f does not change within a ball of radius σ around any point. Such sufficiently small σ allows us to capture the local neighborhoods of the states at this measurement scale on the latent manifold.

Ideally, being able to find $f^{-1} : \mathscr{Y} \to \mathscr{X}$ would suffice to reconstruct the latent domain, but even if f is invertible, it is generally not feasible to identify f^{-1} without access to $\mathscr{X}$. We can, however, try to construct an embedding $\rho : \mathbb{R}^D \to \mathbb{R}^d$ that maps the observations $\{\boldsymbol{y}_i\} \in \mathscr{Y}$ so that the image of $\rho \circ f$ is isometric to $\mathscr{X}$ when σ is known. In our Euclidean setting, it is equivalent to require that the distances in the latent space are preserved, i.e., samples should satisfy $\left\|\rho(\boldsymbol{y}_i) - \rho(\boldsymbol{y}_j)\right\|_2 = \left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|_2$ for any i, j. Following the mathematical derivation in [59], the embedding ρ that preserves the distances should satisfy the condition

$$\frac{1}{\sigma^2} \boldsymbol{C}(\rho(\boldsymbol{Y}_i)) = \boldsymbol{I}, \tag{4}$$

where $\boldsymbol{C}$ is the covariance matrix computed over the embedding of the samples from each burst. In other words, (4) requires whitening the covariance of local samples in the embedding domain.

Implementation: As stated above, our task is to find the embedding function ρ. We use an encoder neural network (E) to learn the embedding function. Following (4), we define the

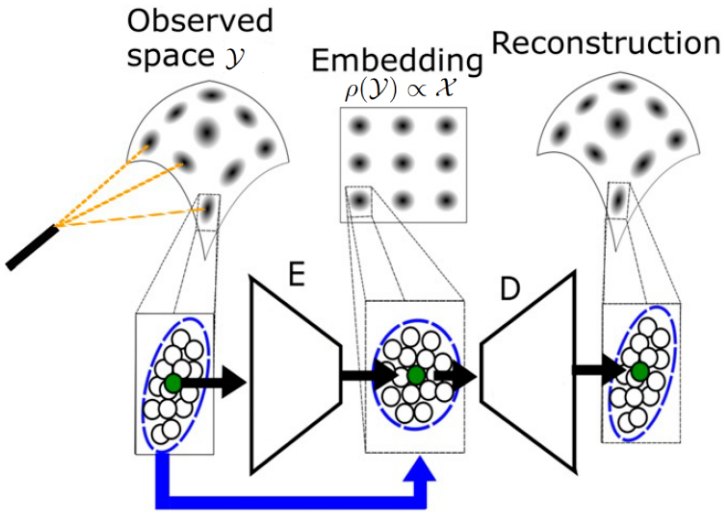


Figure 2.2: An illustration of LOCA adapted to acoustic scene mapping. The observation space $(\mathscr{Y})$ is assumed to model a nonlinear deformation of the inaccessible manifold $(\mathscr{X})$. We attempt to invert the unknown measurement function, utilizing the bursts sampling strategy. LOCA consists of an encoder (E) parameterized by ρ and a decoder (D) parameterized by γ. The autoencoder receives a set of points and corresponding neighborhoods; each neighborhood is depicted as a dark oval point cloud (at the top of the figure), which we implement using a microphone array. At the bottom, we zoom in onto a single anchor point y_i (green) and its corresponding neighborhood Y_i (bounded by a blue ellipsoid). The encoder attempts to whiten each neighborhood in the embedding space, while the decoder aims at reconstructing the input. In practice, a special microphone array will be used to extract bursts of adjacent acoustic samples of RTFs.

loss term to be minimized at each iteration:

$$L_{\text{white}}(\rho) = \frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{\sigma^2}\hat{C}(\rho(\boldsymbol{Y}_i)) - \boldsymbol{I}_d\right\|_F^2, \tag{5}$$

with ρ the learned embedding function up to the current iteration and $\hat{\boldsymbol{C}}(\rho(\boldsymbol{Y}_i))$ is the empirical covariance over a set of M realizations $\{\rho(\boldsymbol{y}_i^{(j)})\}_{j=1}^{M}$. Next, we note that f is an invertible function, and so is its inverse f^{-1}. Since our embedding ρ tries to estimate f^{-1} up to an orthogonal transformation, we can enforce the invertibility of ρ, reducing ambiguity in the solution. The invertibility of ρ means that there exists an inverse mapping $\gamma\colon \mathbb{R}^d \to \mathbb{R}^d$ such that $\boldsymbol{y}_i = \gamma(\rho(\boldsymbol{y}_i))$ for any $1 \le i \le N$. By imposing an invertibility property on ρ, we effectively regularize the solution of ρ away from noninvertible functions. Practically, we impose invertibility by using a decoder neural network (D) parameterized by γ. This approach induces a loss term based on the mean square error (MSE) between samples and their reconstructed counterparts:

$$L_{\text{recon}}(\rho,\gamma) = \frac{1}{N\cdot M}\sum_{i,m=1}^{N,M}\left\|\boldsymbol{y}_i^{(m)} - \gamma(\rho(\boldsymbol{y}_i^{(m)}))\right\|_2^2. \tag{6}$$

The LOCA framework is schematically depicted in Fig. 2.2.

3 Proposed Algorithm - Acoustic Scene Mapping

In the following subsection, we provide complete details about our problem and the proposed solution. To illustrate our studied problem and the proposed algorithm, we provide a visualization of the training scheme in Fig. 3.1.

3.1 Data Collection

In order to implement the burst sampling strategy, we construct a unique microphone array consisting of M components of adjacent microphone pairs. That would enable us to sample the RTFs induced by the sound sources in the same local neighborhood, resulting in bursts of M samples.

The process of collecting data is as follows: a device mounted with the microphone array is free to travel within the acoustic enclosure and pause occasionally. During the pause, each microphone in the array records the sound signals emitted by each sound source. We note that the orientation of the microphone array is unchanged throughout the process. Explicitly, either the device cannot rotate, or it calibrates itself to a predetermined orientation using a compass-based technology or a GPS. The reason is that RTF acoustic features are sensitive to changes in the orientation of the microphone array itself.

Next, we use the recorded signals from each microphone pair to extract the RTF w.r.t each one of the sound sources, using Eq. 3. We use $N_{FFT} = 256$ with 50% overlapping Hamming windows. A single estimated RTF is a complex-valued feature of length $\frac{N_{\text{FFT}}}{2} + 1 = 129$. For working with real-valued deep neural networks (DNNs), we concatenated the real and imaginary parts to form a 258-dimension real-valued feature. Practically speaking, we use only 95 frequency bins for each RTF, as we found out that the best results are achieved when using frequency bins 5-99, corresponding to frequencies 312.5–6190 Hz. If the enclosure includes multiple sound sources, their RTFs are concatenated to create a single long feature vector. We denote the final size of the concatenated feature as D.

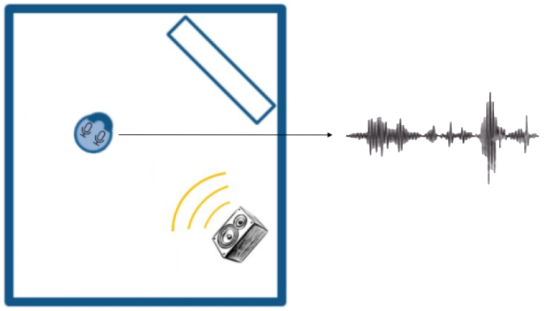
Once we have collected N bursts of samples, each burst consisting of M features, and each feature of length D - we are left with a data tensor of size $[N, M, D]$.

3.2 Training Framework

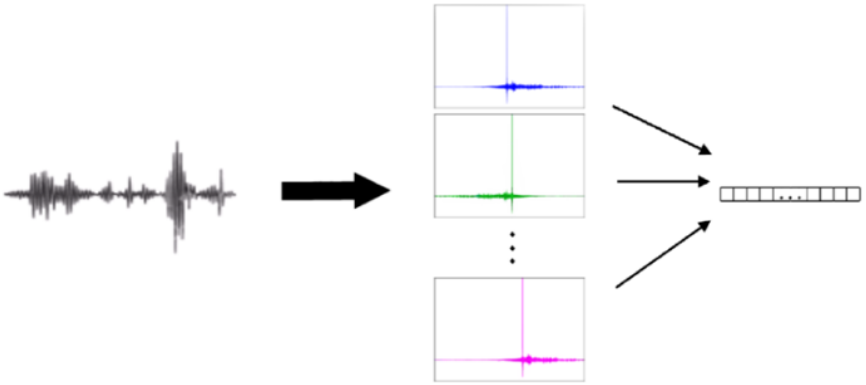
We implement both the encoder and decoder using simple feed-forward architectures. The encoder consists of an initial layer of dimension D with no nonlinearity, followed by five layers of size 200 with 'Leaky-Relu' activations, ending with a latent representation layer of size d. Similarly, the decoder consists of the latent representation layer with no nonlinearity, followed by four layers of size 200 with 'tanh' activations, an additional linear layer of



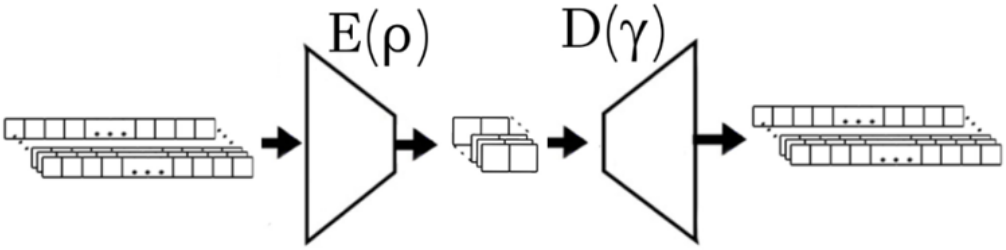
(a) **Problem Statement**: A device mounted with a special microphone array travels in an acoustic enclosure with active sound sources.



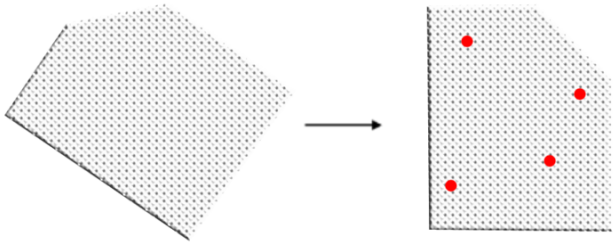
(b) **Step 1 - Data Collection**: The device repeatedly stops to record the signals produced by the sound sources while static.



(c) **Step 2 - Data Preparation**: For each component within the microphone array, RTFs are estimated w.r.t each of the sound sources. The estimated RTFs are then concatenated into a single long vector to create a single data sample.



(d) **Step 3 - Training**: We group together data samples taken from the different components in our microphone array. These sample groups, named bursts, are fed into the LOCA model to learn the encoder whose output is the low dimensional embedding.



(e) **Step 4 - Calibration**: We use a group of anchor points of known location to calibrate the embedding at the output of the encoder with the knowledge we have about the enclosure.

Figure 3.1: A visualization of the problem setup and the training scheme used for acoustic scene mapping.

size 200, and an output reconstruction layer of size D. We use a batch size of 2048 samples and a learning rate of $1e^{-3}$. While the original LOCA formulation applied alternating minimization steps between the whitening and reconstruction loss terms, we found that, in our setting, we can obtain a more stable solution by minimizing a combined loss function. We use 90% of the learning-phase data during training and the rest 10% is used for validation and determining the best model weights. Finally, based on the geometry of the microphone array, we set $\sigma = 6e^{-4}$.

Finally, the model is trained by feeding batches of bursts into the model, taken from the previously described data tensor. Once the training ends and the best weights are loaded, we have an encoder that can generate embedding for our training samples. As explained earlier, the embedding represents the latent manifold up to a rotation and a shift, and therefore the final step would be to find a calibration of the embedding that fits the true coordinates of the scene. Note that up until now, the learning was unsupervised. Now, this is the first time we incorporate our knowledge of the geometry of the problem into the model. To carry out the calibration, we find the affine transformation that achieves the lowest mean absolute error (MAE) with the group of samples of known locations (the anchors).

3.3 Inference Framework

At the inference stage, we have new measurements of positions that the receiver has not seen during training. We can use the learned embedding to carry out inference over these samples and effectively localize the receiver. Inference is done by feeding the new measurements through the encoder and applying the learned affine transformation over the encoder's output. Importantly note that we no longer need the entire microphone array in the inference stage. Instead, we can use only a single component of it. That is a great advantage, as it substantially reduces the required hardware once the acoustic scene has been learned.

4 Experimental Study

We will now introduce the experiments which were done to test our proposed approach.

4.1 Acoustic Scene Mapping in a Simulated Room

Experimental Setup: To implement the burst sampling strategy, we need to construct a microphone array in the same manner discussed earlier in Section 3.1. A single difference from the originally proposed algorithm is that for each component of our complete microphone array, instead of a single microphone pair, we use two perpendicular microphone pairs (a cross of microphones) and concatenate the features of the perpendicular arrays. This change is crucial when working in a two-dimensional environment where movement is allowed along two axes. Consequently, we can describe the burst microphone array as comprising seven

cross-shaped sub-arrays in a circular constellation, each consisting of four omnidirectional microphones. The distance between the microphones in each arm of the cross is 2ℓ, and the radius w.r.t. the constellation's center is r (see Fig. 4.1). For a relatively small radius, this circular array accurately simulates a Gaussian neighborhood sampled in the latent domain, as the burst sampling strategy requires. Practically speaking, ℓ might be larger than r, resulting in overlapping crosses of microphones.

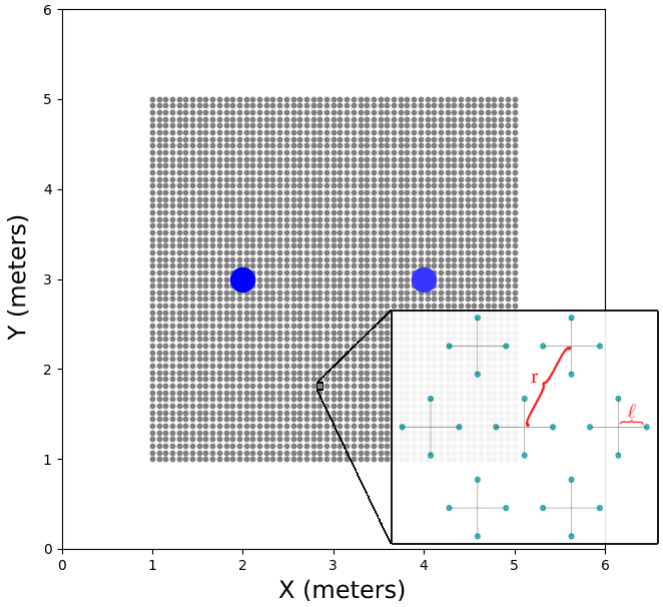


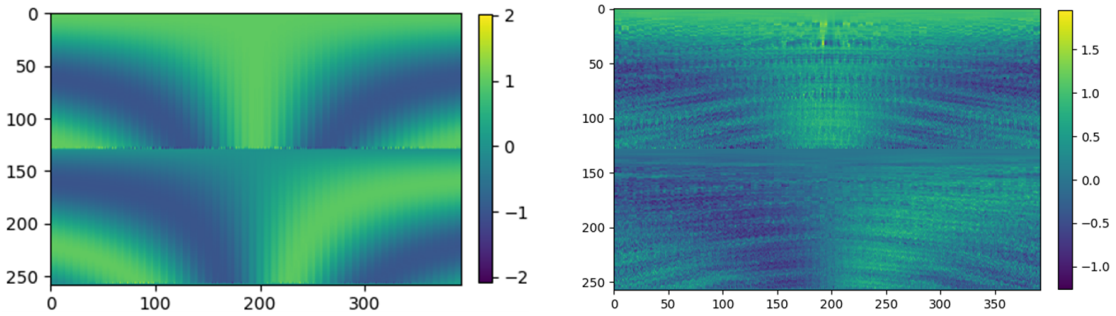
Figure 4.1: Room sampling strategy visualization. Blue circles denote the location of the sound sources. The sampling grid along which the device travels is shown in grey. We zoom in to show the schematic configuration of the burst microphone array: seven cross-like components consisting of vertical and horizontal microphone pairs.

Consider a room of dimensions $[6,6,2.4]$ m. Define an RoI within the room of size 4×4 m, symmetrically positioned around the center of the room, 0.2 m above the room's floor. The following simulation will be divided into two parts: a learning phase and a test phase. In the learning phase, the microphone array is free to move only on a grid with a spatial resolution of 56 samples along each dimension, yielding a total of $N = 3136$ bursts for training and validation, with a spacing of 7.27 cm between the centers of neighboring bursts. In the test phase, we allow the device to travel in the same confined region but not necessarily on the grid. The test set includes 750 bursts from random locations within the RoI. Two omnidirectional sound sources are placed at $[x,y,z] = [2,3,1.7]$ m and $[x,y,z] = [4,3,1.7]$ m, lying 1.5 m above the sampling-grid plane. Note that the positions of the sound sources are unknown to the algorithm. The height of the sound sources above the sampling plane is vital due to simple geometric considerations. It allows phase accumulation between received signals relative to the sources in all directions. In addition, it became apparent by a series of experiments that using only a single emitting source may produce heavily deformed embeddings at the edges far from the source. For that reason, we used two sources in the

simulations. Note that the sound sources are not simultaneously active, thus allowing for the estimation of the individual RTFs. A top-view visualization of the sampling constellation is shown in Fig. 4.1.

Reverberated Data and Training: In order to generate the reverberant acoustic data, we used the GPU-accelerated RIR Simulator [60, 61]. To create the learning-phase data, the synthetic RIRs, $a_i(n)$, were convolved with 5−second long white noise signals, $s(n)$. The acoustic data at the test phase is simulated by convolving the RIRs, $a_i(n)$, with 8−second long speech signals, $s(n)$, drawn from the TIMIT dataset [62]. We examined three reverberation times ($\mathrm{RT}_{60} = 160, 360, 610$ ms), set the sampling frequency to $f_s = 16$ kHz, and the speed of sound to $c = 343$m/s. We also set the microphone array parameters to $r = 2$ cm and $\ell = 3$ cm. Using this configuration, at any given location and for each source, we can estimate the horizontal RTFs and the vertical RTFs, totaling seven horizontal and seven vertical RTFs for a single burst.

Construction of the data tensor: we take 95 bins from each complex RTF for each single sound source to construct a real-valued feature of length 190. Since we have both vertical and horizontal RTF components in our burst, we concatenate them to obtain a feature vector of length 380. Finally, having two speech sources, we repeat the process and concatenate both results. We eventually end up with a data tensor of shape $[N, M, D] = [3136, 7, 760]$. In terms of the LOCA terminology, we can write that $\mathscr{Y} \subset \mathbb{R}^{760}$, and we wish to reconstruct a 2-D latent embedding, $\mathscr{X} \subset \mathbb{R}^2$. Fig. 4.2 gives a visual inspection of our acoustic features, comparing between the moderately reverberated case and a clean case with no reverberation at all.



(a) No reverb. (b) $\mathrm{RT}_{60} = 360$ ms.

Figure 4.2: A visualization of the RTF input features corresponding to the vertical microphone pair components of the leftmost sampling grid-line, for $\mathrm{RT}_{60} = 360$ ms. A line in the grid consists of 56 bursts, each containing $M = 7$ RTFs, yielding 392 RTFs. Each group of 7 consecutive columns in the figure belongs to a different burst, appearing according to their order along the line. The data consists of an upper and a lower block, corresponding to the real and the imaginary parts of the RTFs, respectively.

Simulation Results:

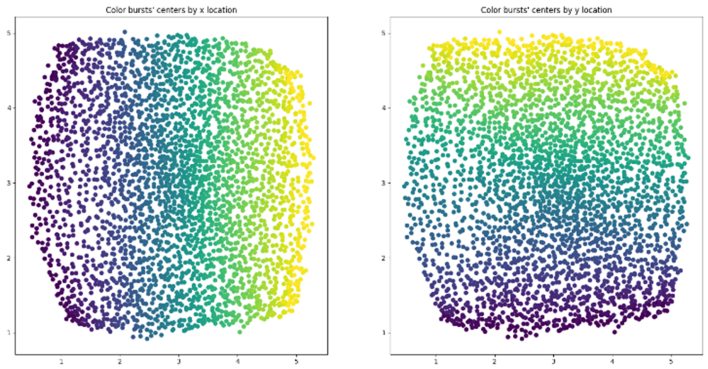
As discussed earlier, our embeddings are invariant to shifts and orthogonal transforma-

tions; therefore, they should be calibrated using a set of anchor points. The calibration process is the first time we incorporate our knowledge of the true geometry of the problem into the model. In Fig. 4.3, we visualize the calibrated embedding provided by our framework. Our learned embedding demonstrates a high correlation between the main directions of the embedding and the true $x-y$ axes characterizing the square sampling grid. While increasing the reverberation level may deform the embedding, the model can still capture the directional correlation and reveal the square shape of the grid (see Fig. 4.3). We suspect the reverberation-attributed degradation stems from the LOCA's assumption that f is a smooth function. In high reverberation, small perturbations in the location translate to relatively sharp changes in the RTF, leading to less smooth f. As the high-frequency bins tend to change rapidly with the burst's position, we empirically verified that we should discard these bins to achieve better results.

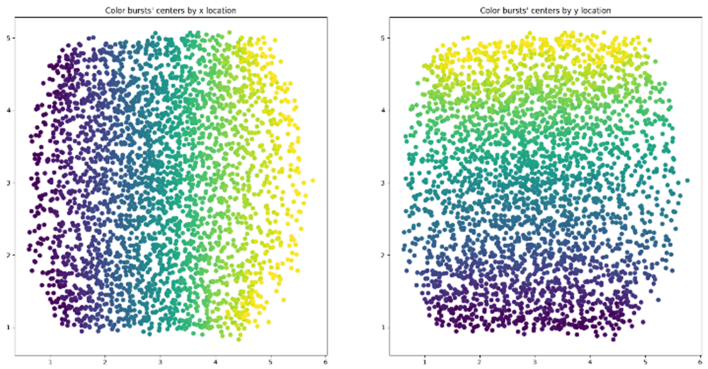
Comparison with Manifold Learning Methods: Next, we compare the results with other baseline methods. Since we focus on unsupervised acoustic scene mapping, we compare LOCA to alternative dimensionality reduction methods. Specifically, we compare Principal-Component-Analysis (PCA), Metric-Multi-Dimensional-Scaling (MMDS) [63], diffusion maps (DM) [64], and anisotropic diffusion maps (A-DM) [58]. The above methods are commonly used in manifold learning applications and are justified by our assumption of the existence of the acoustic manifold. Hence, they may serve as valid candidates for comparison. We stress that these schemes are tested with the same data used to test LOCA.

As discussed earlier, we quantify the performance using the test-phase data. We calculate the mean absolute error (MAE) of the distance between the positions in the calibrated embedding space and their matching counterparts in the latent space. Note that we first calibrate the embedding using an orthogonal transformation and a shift before calculating the MAE. Additionally, in DM and A-DM, where we need to carry out hyper-parameter tuning for σ, we initiialy find the optimal σ value by selecting the embedding that leads to the minimal MAE, given a small set of four anchors. In Table 4.1, we compare the performance measures of all methods for the different reverberation times. It is evident that LOCA leads to the best results in terms of MAE for all reverberation levels. Finally, we should emphasize the advantage of LOCA over the other methods regarding inference time. Once we have constructed a mapping of the RoI in the training phase, we can perform self-localization using new samples, where now we can use only a single cross instead of an entire burst. LOCA presents a fast and simple inference process based on the DNN's forward pass. In contrast, Kernel-based methods require calculating the kernels between the new and all other training samples, which is a cumbersome process. To quantify this gap, we compared the inference time of our method with kernel schemes and observed that we could reduce the run time by four orders of magnitude.

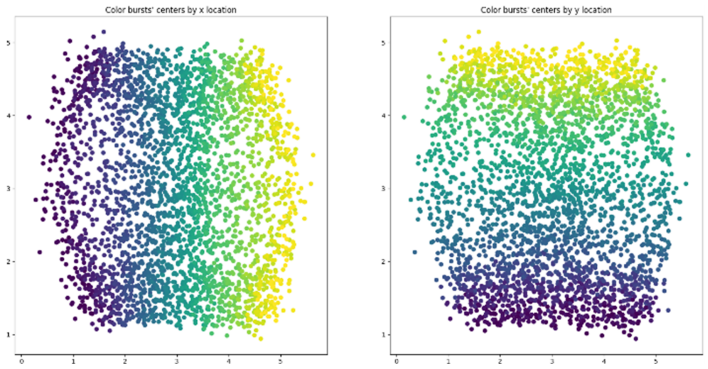
For the following demonstrations, we will be focusing on the low reverberated case of $\mathrm{RT}_{60} = 160$ ms, where the results of the different dimensionality reduction schemes are



(a) $RT_{60} = 160$ ms



(b) $RT_{60} = 360$ ms

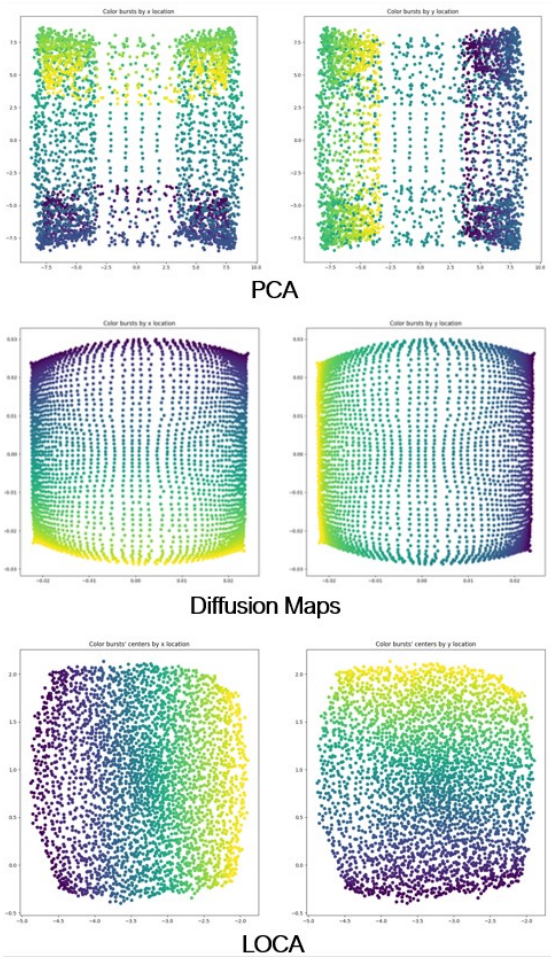


(c) $RT_{60} = 610$ ms

Figure 4.3: 2-D geometric reconstruction of the training net grid achieved by the transformed embedding of our framework (LOCA). The coloring indicates the correlation with the true coordinates of the original scene where the left and right panels show the correlation with X and Y coordinates respectively.

Table 4.1: MAE (cm) of reconstruction using test data.

Method	160 ms	360 ms	610 ms
PCA	72.2	72.5	74.3
MMDS	73.4	79.1	82.5
DM	20.4	22.7	65.1
A-DM	17.8	18.3	33.8
LOCA	**11.3**	**13.4**	**18.5**



Method \ RT_{60}	160 ms	360 ms	610 ms
PCA	75.8	74.7	75.9
MMDS	73.3	79.5	82.8
DM	26.2	23.6	69.8
A-DM	18.4	25.4	33.8
LOCA	**13.1**	**14.9**	**18.7**

Table 1: MAE (cm) comparison of the embeddings gained by each method.

Figure 4.4: A comparison between the embeddings achieved for the training net grid by all methods. Similarly to Fig. 4.3 the correlation with the true X and Y coordinates is visualized.

best comparable to LOCA's result. We begin with a visual comparison of the embeddings achieved by the different methods in Fig 4.4. Note that the embeddings here are attributed to the original training net-grid, resulting in a grid pattern visible in most embeddings. As visible, both DM and A-DM do a great job at capturing the orientation and ordering of the grid, however, they struggle with the grid points at the far edges, attributing them much greater density than their actual density. On the opposite hand, in LOCA the grid pattern is less visible, but it is more reliable at preserving the actual densities and distances between points. We further demonstrate LOCA's superiority at preserving the distances in the isometry measure graph in Fig. 4.5. The isometry measure graph is a scatter plot representing the embedding distances vs. the actual distances for each method. In order to create the graph, we randomly picked 3000 pairs of points from the pool of test data samples, and compared their embedding distances to their actual distances. A true isometry would preserve distances and behave similarly to the black dashed line. As can be visually inspected, LOCA is the closest to a true isometry.

Comparison with a Classical Approach: To complete the study, we have tested a naïve

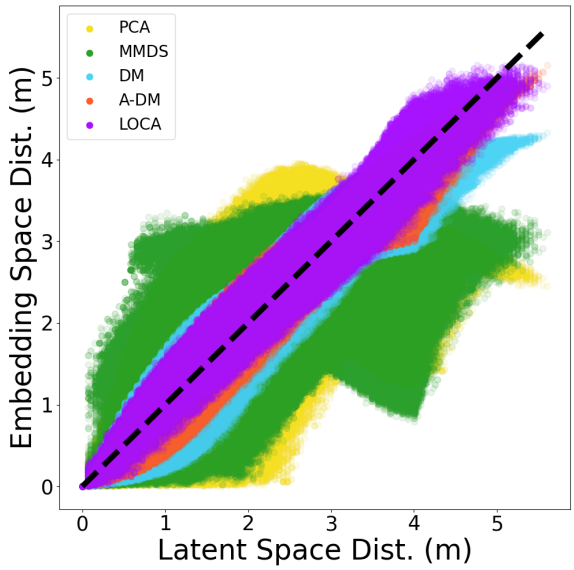


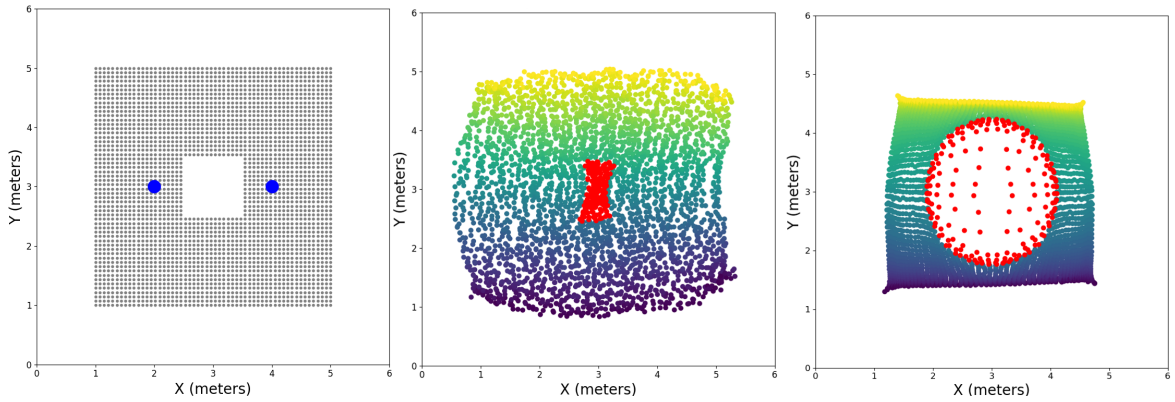
Figure 4.5: A demonstration of the isometry measure of the different methods at the case of $\mathrm{RT}_{60} = 160$ ms. The black dashed represents a perfect isometry, and the better methods are the ones whose embedding space distances are tighter to it.

classical approach for acoustic scene mapping of the discussed RoI. We considered a coarser grid than before, consisting of 10×10 samples, with a spacing of 44.4 cm between neighboring locations. To further simplify the classic approach, the sound sources were lowered to the grid plane. For this approach, which is further explained in Section 3.6 of [65], we used two crosses of microphone pairs identical to the central cross array in Fig. 4.1. One is a static reference array located in the middle of the RoI, whereas the second array moves successively along the grid points. We then used GCC-PHAT to estimate the TDOAs between the moving array and the reference array centroid w.r.t. each speaker. Additionally, we used SRP-PHAT [15] to estimate the DOA of each speaker w.r.t. each array. This scheme is sensitive to the reverberation level, with a significant performance drop in MAE from 13.7 cm for $\mathrm{RT}_{60} = 160$ ms to 64.3 cm for $\mathrm{RT}_{60} = 360$ ms and down to 74.8 cm for $\mathrm{RT}_{60} = 610$ ms.

4.2 Out-of-Sample Extrapolation Capabilities

Extrapolation Capabilities: We now examine the extrapolation capabilities of the proposed scheme and compare them to the A-DM manifold learning scheme. For that purpose, we use the same experimental setup as before with a few modifications. We still use the region between $[1,5] \times [1,5]$ m to learn the manifold. However, the data in the center of the room, originating from the region between $[2.5,3.5] \times [2.5,3.5]$ m, is unavailable during the training phase. The described scenario is depicted in Fig. 4.6a. For this demonstration, we consider the case of $\mathrm{RT}_{60} = 160$ ms. As evident in Fig. 4.6b, LOCA maintains good manifold reconstruction, indicated by the uniform distribution of the embedding. Note that the extrapolated region also spreads out proportionally well. On the other hand, A-DM strug-

gles to learn the latent manifold when a big slice of the data is unavailable. That is clearly visible from Fig. 4.6c, where it is apparent that the manifold is exceedingly stretched, and the extrapolated region becomes disproportionally large. Both the visual inspection and the reported MAE values confirm that LOCA's ability to extrapolate outperforms that of the older manifold learning method.



(a) Training region (b) LOCA (c) A-DM

Figure 4.6: A visualization of the extrapolation capabilities of LOCA and A-DM for $\mathrm{RT}_{60} = 160$ ms. The samples from the extrapolated region are shown in red. The MAE values of the extrapolated region are 16.1 cm for LOCA and 67.4 cm for A-DM.

Part II
One-Dimensional Tracking

5 Problem Formulation

In the current section we formalize the problem and introduce the instantaneous relative transfer functions (iRTFs) - the acoustic features we use to perform tracking.

5.1 Problem Statement

Consider an acoustic enclosure where a dynamic speaker is moving and recorded by a pair of microphones. Following previous works [49, 52], we focus on the one-dimensional setting, where the speaker can only travel on an arc of radius R around a microphone pair. In this setting, the DOA, $\theta(t)$, represents the time-varying angular displacement of the speaker along the arc path around the microphone pair. We also assume that we are given a small set of anchor points, i.e., samples whose positions are known and labeled. The discussed setting is illustrated in Fig. 5.1.

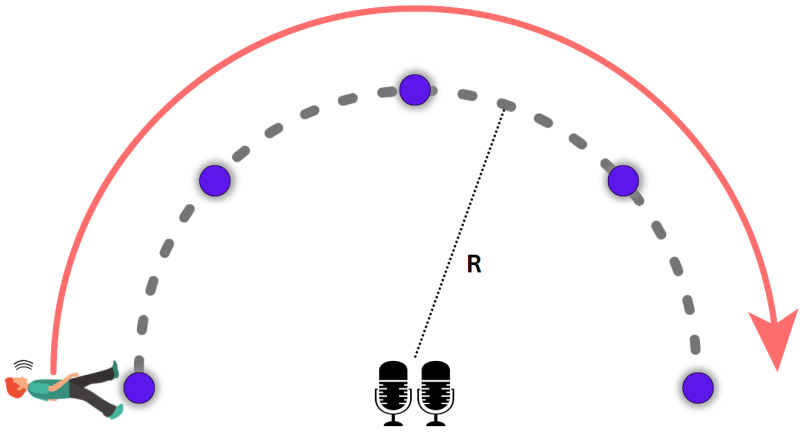


Figure 5.1: An illustration of the experimental setup. Dynamic speakers are confined to an arc of radius R around a pair of microphones, both in the training and test phases. $0 \leq \theta(t) \leq 180$. Blue circles denote labeled examples.

5.2 Instantaneous Acoustic Features

Consider time-domain acoustic measurements of the form

$$z_i(t) = \{a_i * s\}(t) + u_i(t), \tag{7}$$

with $s(t)$ being the source signal, $a_i(t)$, the unknown RIRs relating the source and each of the microphones, $i = \{1,2\}$, and $u_i(t)$ noise signals independent of the source. Note that, in our case, the speakers are moving. Consequently, the RTF is time-varying, and using Eq. 3 will no longer give an estimation for the local RTF at a given position. Instead, we will use a set of N_c context time frames to estimate the iRTF, which approximates the local RTF. In our formulation, the parameter N_c is assigned even values.

Assuming negligible noise and selecting $z_1(n)$ as the reference signal, the iRTF, $H(t,k)$, can be estimated using:

$$H(t,k) \approx \frac{\hat{S}_{z_2 z_1}(t,k)}{\hat{S}_{z_1 z_1}(t,k)} \approx \frac{\sum_{n=t-\frac{N_c}{2}}^{t+\frac{N_c}{2}} Z_2(n,k) Z_1^*(n,k)}{\sum_{n=t-\frac{N_c}{2}}^{t+\frac{N_c}{2}} Z_1(n,k) Z_1^*(n,k)}, \tag{8}$$

where $\hat{S}_{z_1 z_1}(t,k)$ and $\hat{S}_{z_2 z_1}(t,k)$ are the power spectral density (PSD) and the cross-PSD at time t, respectively, and $Z_i(n,k)$ are the STFT domain representations of the measured signals for $i = \{1,2\}$. Ultimately, we focus on a portion of the frequency bins range, namely $k_1, \ldots, k_P$, that includes most of the spectral energy. That would allow for a reliable estimation of the acoustic features, further improving the localization capability. We finally define our feature vector at time t as the concatenation of the real and imaginary parts of the iRTFs in the limited frequency range:

$$\begin{aligned} \hat{\mathbf{H}}(t) = [&\mathrm{Re}\left(H(t,k_1), \ldots, H(t,k_P)\right), \\ &\mathrm{Im}\left(H(t,k_1), \ldots, H(t,k_P)\right)]. \end{aligned} \tag{9}$$

The Relation Between the iRTF Features and the Acoustic Manifold:

Following the discussion in Sec. 2.2 regarding the acoustic manifold, we recall that the RTF features are points that lie on the low dimensional manifold, which are represented in high dimensional space. The manifold itself is controlled only by a set of small latent parameters, in our case that would be simply the DOA of the speaker.

In practice, we consider the iRTFs as our acoustic features, which are merely an approximation of the RTFs.Consequently, the instantaneous features might lie beside the theoretical acoustic manifold, and we get an *almost* locally linear manifold. However, it *can* be considered linear in the local neighborhoods for all practical purposes. This reasoning motivates us to apply a manifold learning technique that utilizes the local linearity property of the manifold, namely the LOCA framework.

6 Proposed Approach

6.1 Modified LOCA

Following the introduction of the LOCA dimensionality reduction model in Sec. 2.3, we aim to introduce a modified and improved LOCA model. The methodological adjustments we implement further optimize the learning process, allowing for faster convergence and, most importantly, stabilizing the learning process by avoiding convergence to local minima.

The first change we include is the addition of a new anchoring loss term. The idea is to stabilize the solution and remove arbitrary scaling and rotations from the embedding learned

by LOCA. Since we learn an embedding ρ that is only isometric to the inverse measurement function f^{-1}, we need to relate the embedded samples - i.e., the predicted speaker positions, to their real-world coordinates. In Sec. 3 and in [59], the incorporation of real-world coordinates into the embedding was done only past the training stage, where an orthogonal transformation for the embedding was found by comparing the embedded anchor samples to their true positions. However, we found that using a small set of anchor points during training would be sufficient to accurately calibrate the embedding and improve the training process's overall stability. Assume that we have a set of N_L labeled training RTF samples, $\{x_k\}_{k=1}^{N_L}$, whose positions, $\{\theta_k\}_{k=1}^{N_L}$, are known. We define the anchoring loss term as the mean square error between the embedded samples and their true positions

$$L_{\text{anchor}}(\rho) = \frac{1}{K}\sum_{k=1}^{N_L} |\rho(x_k) - \theta_k| \,. \tag{10}$$

Finally, our objective is to minimize the combined loss term

$$L = \lambda_{white} \cdot L_{\text{white}} + \lambda_{recon} \cdot L_{\text{recon}} + \lambda_{anchor} \cdot L_{\text{anchor}} \tag{11}$$

where λ_{white}, λ_{recon} and λ_{anchor} are the weighting factors for the whitening, the reconstruction and the anchoring loss terms respectively.

The second adjustment we implement was already adopted and introduced in Sec. 3.2. While the original work suggests a step-by-step minimization of the loss terms, we minimize them mutually, resulting in a much more stable learning process. In section 7.6 we provide an ablation study to assess the impact of the two discussed methodological changes.

The entire training scheme for the modified LOCA framework applied to one-dimensional speaker tracking is illustrated in Fig. 6.1.

6.2 Synthetic Aperture Paradigm for Bursts Collection

In order to use LOCA to learn an embedding that is isometric to the latent space, it is necessary to be able to sample bursts, or clusters, of neighboring samples. However, acquiring such data can be challenging. In a previous study that involved performing a static 2-dimensional acoustic-scene mapping [1], the authors devised a unique, albeit complex, microphone array that allowed for the collection of bursts of adjacent samples. Fortunately, when we limit ourselves to the 1-dimensional setting, we can take advantage of the dynamics of the speakers. When recording a moving speaker along the arc, we can think of measurements of consecutive instantaneous acoustic features as if they arrive from a local neighborhood in which the measurement function does not vary dramatically. This allows us to define local bursts. In the following paragraphs, we will further formalize this concept.

Given recording, $r(t)$, of a dynamic speaker that is confined to a 1-dimensional arc, our goal is to collect adjacent samples from the recording to train our network. The recording

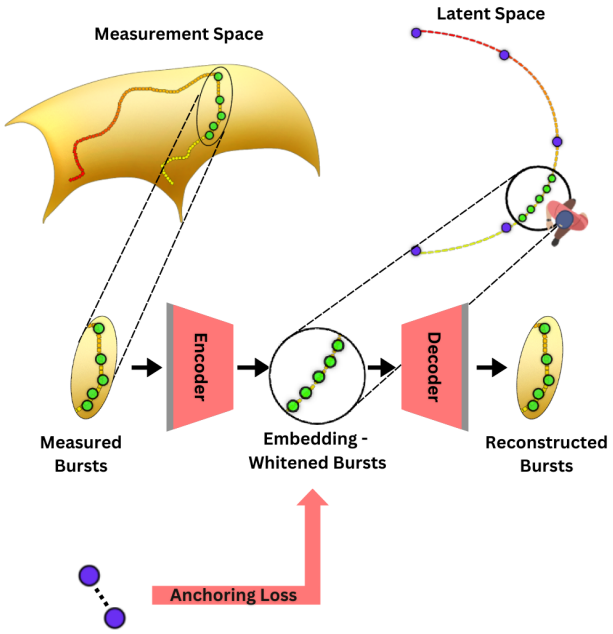


Figure 6.1: An illustration of the modified LOCA framework training scheme. The latent space is a low-dimensional manifold that resides in a high-dimensional measurement space. Bursts of adjacent samples from the measurement space are fed into the encoder, aiming to whiten their embedding (see Eq. 5). The embeddings of the anchors (blue circles) are compared to their true positions via the anchoring loss (Eq. 10). The embeddings are then fed into the decoder to reconstruct the input (Eq. 6).

consists of two channels - $\{z_1(t), z_2(t)\}$. To ensure the reliability of the acoustic features we extract, we need to collect trustworthy samples. We cannot train our network with samples taken from speech segments that are empty or lack content in certain frequencies. To estimate the iRTF features accurately, it is essential to use recording segments that have sufficient active speech. To ensure this, the first step is to apply a voice activity detector (VAD) algorithm that identifies and keeps only the active time frames. Here, we use a window size of $N_{FFT} = 2048$ and overlap of 75% between the adjacent windows to divide the recording into a group of time frames, $\{t_i\}_i$. Once the VAD algorithm is applied to the time frames, we are left with only the active time frames, enabling us to estimate the iRTF features accurately,

$$T = \{t_i \big| \text{VAD}\,(r\,(t_i)) = 1\}. \tag{12}$$

Let us define N_c^{train} as the number of context frames required to estimate the iRTFs during the training phase. We can also define N_c^{test} as the number of context frames needed for estimation during the test phase. However, in general their value might differ, and it is up to the user to decide the number of context frames used during the test as it is closely related to the inference latency. To ensure that the estimation of training samples is done using sufficient active speech, we aim to keep only those time frames that have an active range of

context frames (time segment):

$$\hat{T} = \left\{ t_i \in T \middle| \begin{array}{c} \forall t_i - \dfrac{N_c^{train}}{2} \leq t_j \leq t_i + \dfrac{N_c^{train}}{2}, \\ \text{VAD}\left(r\left(t_j\right)\right) = 1 \end{array} \right\}. \tag{13}$$

After collecting all of the active time segments, our goal is to create adjacent bursts of iRTF samples. As we discussed earlier, we can consider consecutive iRTF estimations as measurements that arrive from adjacent positions on the arc by utilizing the movement of the speaker. We set the size of each burst as $M = 5$, and collect all groups of 5 time frames in $\hat{T}$ with a spacing of 2 time-frames to construct our training set. The time-frame bursts are of the form

$$\tilde{T}_i = \{t_i - 4,\ t_i - 2,\ t_i,\ t_i + 2,\ t_i + 4\}. \tag{14}$$

Finally, we use each of the collected time-frame bursts in $\{\tilde{T}_i\}_i$ and estimate its iRTFs using Eq. 8 over each time frame. The achieved results are bursts of iRTF samples $\{\boldsymbol{x}_i^{(j)}\}_{j=1}^{M}$, where each burst contains $M = 5$ samples of dimension D. Since our primary focus is speech data, we consider only the frequency-bin range of $[41 - 320]$ which corresponds to the frequencies 312–2500 Hz for our choice of $N_{FFT} = 2048$. This range includes most of the speech power distribution. Given that we are required to concatenate both the real and imaginary parts of the iRTF (Eq.9), we are left with features of dimension $D = 560$. Algorithm 1 outlines the process of training-bursts collection and is followed by Algorithm 2 which concludes both the training process and the tracking methodology.

We note that the essential requirement for the sample bursts defined in Section 2.3 is that the latent states associated with each burst are independent and identically distributed (i.i.d.) Gaussian variables. In simpler terms, this means that the locally spatial distributions of samples within bursts behave similarly and follow an isotropic Gaussian distribution. Note, however, that in the case where $\sigma \ll 1$, other isotropic distributions could also be used to model the local perturbations around states. Fortunately, having speakers that move at a constant velocity, we can set the size of each burst as $M = 5$ and gather five evenly spaced samples in time. It also follows that they are uniformly spaced in the latent space, thus satisfying the required conditions. A similar concept, where movement is utilized to increase the resolution of the measured signal, is known in radio detection and ranging (RADAR) systems as the synthetic aperture paradigm. In our case, it is not a signal whose resolution we wish to increase but the sampling of the acoustic manifold itself. Essentially, the speaker's movement along the path while the microphones sample the acoustic manifold in local neighborhoods is comparable to moving an antenna to increase the spatial resolution of a RADAR image. You can see a conceptual depiction of this paradigm in Fig. 6.2.

Algorithm 1 Training-Bursts Collection

Input: Two-channeled training recordings, $r(t) = \{z_1(t), z_2(t)\}$, of dynamic speakers confined to the 1-dimensional arc.
Output: Tensor of dimensions $[N, M, D]$, containing N bursts of M local iRTF samples, each of dimension D.

for each train recording $r(t)$ **do**
1. Apply a VAD to find which time-frames t_i contain active speech (see Eq. 12).
2. Find the group $\hat{T}$ which contains only the time frames $t_i \in T$ whose entire set of context frames is active (see Eq. 13).
3. Collect time-frame bursts - all groups of $M = 5$ time-frames in $\hat{T}$ with spacing of 2 time-frames (see Eq. 14).
4. According to equations (8, 9), estimate the D-dimensional iRTFs, $\hat{\mathbf{H}}(t)$, for each time-frame t in each time-frame burst $\tilde{T}_j$. Store the iRTF-bursts in the output tensor.

6.3 Model Training and Parameters

We implemented both the encoder and decoder using simple feed-forward architectures. Similarly to Sec. 3.2, the encoder consists of an input layer with no nonlinearity, which is followed by five consecutive layers of size 200 with 'Leaky-Relu' activations, ending with a latent representation layer. This time, the input layer is of dimension $D = 560$, and the latent embedding is of size $d = 1$. Similarly, the decoder consists of the latent representation layer with no nonlinearity, followed by four layers of size 200 with 'tanh' activations, an additional linear layer of size 200, and an output reconstruction layer of size D. We use batches of size 2048 samples and a learning rate of 10^{-3}. The value of σ is set to $6 \cdot 10^{-4}$. In the original manuscript of LOCA, the authors apply alternating minimization steps between the whitening and reconstruction loss terms. We found, however, that a more stable solution is obtained by single-step minimization of the combined loss function in (11). The loss terms are weighted using $\lambda_{white} = 1$, $\lambda_{white} = 25$ and $\lambda_{anchor} = 100$. Finally, we use 90% of the learning-phase data during training, and the rest 10% are used for validation and determining the best model weights.

Algorithm 2 Synthetic Aperture LOCA for Semi-Supervised Speaker Tracking

Training Stage

Input: N_L labeled samples, synthetic aperture bursts tensor of dimensions $[N, M, D]$.

Output: An encoder network (E) parameterized by ρ

for N_{epochs} **do**

1. Feed batches of bursts into the model, single-step minimizing the loss function in (11).

Test Stage - Tracking

Input A new recording containing a pair of signal measurements, $r^*(t) = \{z_1^*(t), z_2^*(t)\}$, produced by an unknown source moving in an unknown location.

Output Estimated DOA $\theta^*(t)$

1. According to equations (8, 9) estimate the corresponding iRTF, $\hat{\mathbf{H}}(t)$, at each time frame t, using the context frames.

2. predict the DOA using $\theta^*(t) = \rho(\hat{\mathbf{H}}(t))$

7 Experimental Study

In this section, we carry out several experiments to compare the performance of our method with that of other schemes. We highlight the merits and limitations of our approach. Our experiments involve acoustic settings with reverberation levels of up to $RT_{60} = 600$ ms, and we consider various spatial signal to noise ratio (SNR) conditions of either 30, 20, or 15 dB. The distance between our microphones is set to 8 cm, and the radius of the arc is set to 2 m. The evaluation and comparison of all of the tracking methods would focus on the broadside of the microphone configuration, considering DOAs in our range of interest $[30^\circ - 150^\circ]$.

7.1 Evaluation and Performance Measures

To better understand the performance of all methods, we evaluate them using two metrics. We define outlier DOA estimations as instances that diverge more than 20° from the true DOA. Our first performance measure is, therefore, the accuracy of estimation, which is simply the percentage of inlier estimations.

$$\text{Accuracy} = \frac{|\{i | 1 \leq i \leq N_{test}, \quad |\theta_i^* - \theta_i| \leq 20^\circ\}|}{N_{test}} \ \% \tag{15}$$

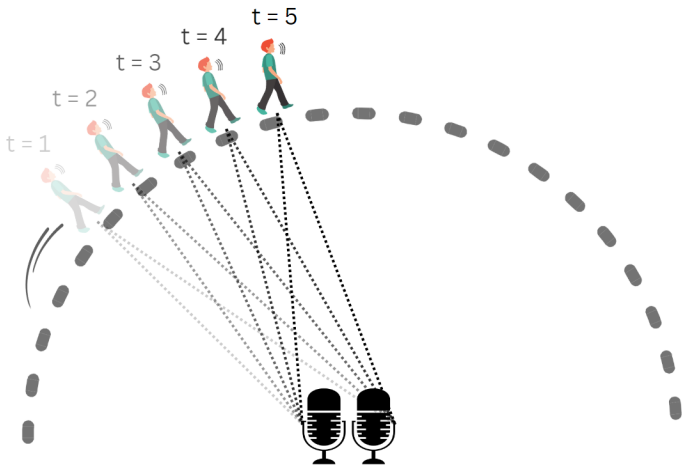


Figure 6.2: An illustration of the synthetic aperture paradigm used for collecting bursts of measurements. Samples recorded at consecutive time-frames are grouped together to form the sample bursts $\{y_i\}_{i=1}^{M}$ for $M = 5$. For a speaker moving at a constant velocity, the latent states attributed to these samples, $\{x_i\}_{i=1}^{M}$, follow an isotropic 1-dimensional uniform distribution in the latent space, thus satisfying the requirements for using the LOCA framework.

The second performance measure would be the root mean square error (RMSE) of estimation. Note that only inlier estimations are considered for calculating the RMSE over the test set.

$$\text{RMSE} = \sqrt{\frac{1}{N_{test}} \sum_{n=1}^{N_{test}} \left(\theta_i^* - \theta_i\right)^2} \tag{16}$$

The test set used will be described in the context of each experiment.

7.2 Simulation I - Method Comparison

In the following simulation, we compare our method with a group of competing algorithms. As stated before, we present the results for different acoustic settings, each with a different level of interference noise and reverberation, examining different amounts of anchor samples. For Each acoustic setting, a Monte Carlo simulation is conducted. A single simulation comprises 20 different scenarios that are determined by the position of the microphone array within a $[6, 6, 2.4]$ m simulated room and the random orientation of the arc. In this simulation, for each scenario, a LOCA-neural network is trained and evaluated on the data of the scenario itself. Consequently, a distribution of the RMSEs over the different scenarios is obtained, together with their average accuracy.

7.2.1 Training Data

To create the training dataset, 28 random speakers were sampled from the TIMIT dataset [62], equally distributed between males and females. Several recordings from each speaker were concatenated to create an extended recording of continuous speech. To simulate the

moving speakers' signals we used the signal generator package[1] for MATLAB. For each scenario, the extended recordings of each speaker were used to simulate a single full traversal over the arc at a constant velocity of 0.25 m/s.

The interference signals, $u_i(t)$, are generated using a simulated white Gaussian noise (WGN) source located at $[4, 1, 1.8]$. The source is located 60 cm above the plane where the microphones are stationed, at $z = 1.2$ m. These signals are attributed to a static speaker and are therefore generated by using the RIR generator[2] to simulate their RIRs, and then convolving the RIRs with WGN signals.

The 28 obtained two-channeled signals of each scenario are then used along with Algorithm 1 to collect all possible bursts for training, resulting in $117,000$ training bursts. The VAD algorithm that is used during this process is the WebRTC-VAD[3], with the aggressiveness level set to 2. The amount of context frames used to estimate the training iRTFs is $N_c^{train} = 10$.

In this simulation, we are exploring the scenario where only one speaker is used to provide labeled anchor samples. Specifically, we are investigating cases where the number of anchor samples N_L is 3, 4, or 5, and the anchors are evenly placed along the arc.

An additional train dataset was created in order to evaluate LOCA's performance when it is trained using a dynamic WGN source signal which traverses the arc only once. Here, a single train recording is simulated in the same manner as the TIMIT recordings. After applying Algorithm 1, we are left with 1520 training-bursts only. The idea here is to exploit the fact that the power spectral density of the WGN is richer. This approach might be advantageous in easing the training process - requiring only a single traversal of the sound source over the arc, thus resulting in substantially fewer training bursts and much faster model convergence. The drawback, however, is that the learned model should be able to generalize to our test set, which consist of speech data. In the result section, the two approaches are differentiated by using either of the terms *LOCA_Speech* or *LOCA_WGN*.

7.2.2 Test Data

For the test dataset, an additional 28 new speakers were sampled from the TIMIT dataset. The signals were simulated in the same methodology as in the training dataset, with a single difference that the test speakers' velocities vary between the values $[0.25, 0.55, 0.85, 1.15]$ m/s. For each speaker, 20 test samples are drawn from the region of interest per test velocity, resulting in an overall of $2,240$ test samples for each scenario. In this simulation, the number of context frames used to estimate the test iRTFs is $N_c^{test} = 20$.

[1] Available at the repository - github.com/ehabets/Signal-Generator
[2] Available at the repository github.com/ehabets/RIR-Generator
[3] Available at the repository - https://github.com/wiseman/py-webrtcvad

7.2.3 Competing Methods

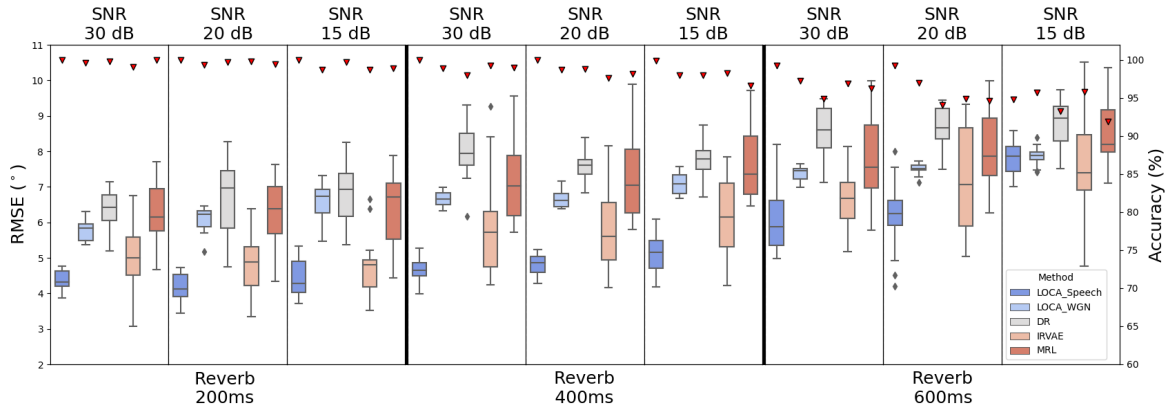
The competing methods we consider are deep ranking (DR) [52], manifold regularization learning (MRL) [49] and IRVAE [66]. It is important to acknowledge that while both DR and MRL were adopted and evaluated for the specific aim of speaker localization and tracking, IRVAE has not been applied to acoustic data thus far. A single challenge with IRVAE was that while it learns an isometric mapping for the embedding, the authors don't provide a means to anchor the embedding coordinates to the real-world positions. Our most effective solution involved post-training calibration of the embedding using the anchor set, similar to the calibration in [1].

7.2.4 Results and Discussion

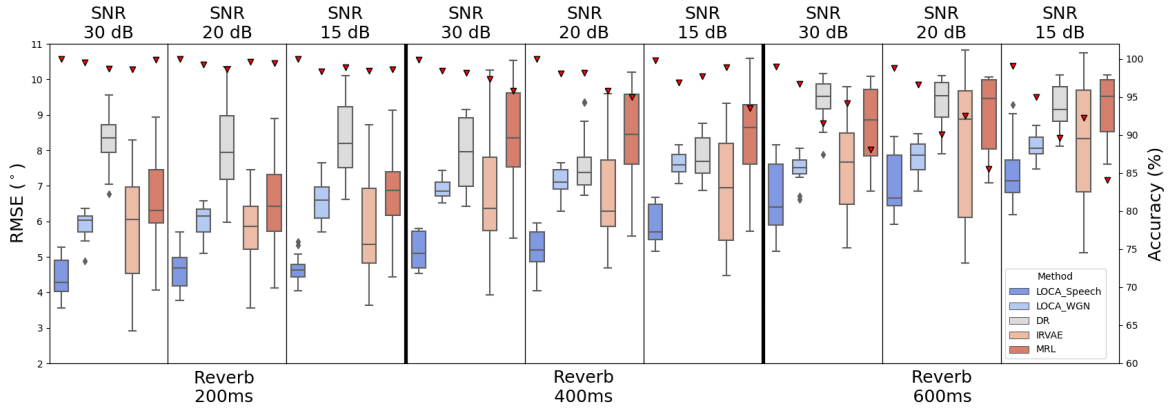
Fig. 7.1 includes multiple box plots that depict the performance of the different methods over the different acoustic settings, considering the different amounts of anchor samples. Examining *LOCA_Speech*, it is evident that we obtain very impressive results, having reached median-RMSEs as low as 4° with an accuracy of nearly 100 %. Even though the training is performed on fixed velocity sources, the results indicate that all of the methods are able to generalize their predictions to the higher speeds as well. We stress that a minimal amount of labeled anchor samples was required in order to achieve this notable performance. It is only under extreme conditions, where both the reverberation and the interference intensity are high, that we see a noticeable decrease in performance compared to the other methods. We note that at first glance it may seem that sometimes other methods, namely the IRVAE, obtain better RMSEs. In most acoustic settings, however, *LOCA_speech* provides the best performance in terms of accuracy while more importantly, judging the whiskers length, also provides more reliable embeddings. In that sense, IRVAE, which might appear as a good approach, is very inconsistent. In the case of *LOCA_WGN*, we observe highly stable results, indicated by extremely thin whiskers. However, the performance measures are not as robust as those achieved with *LOCA_Speech*. This confirms our concern that the learned model might struggle to generalize from the WGN-induced training data to the dynamic-speech-induced test data. Nevertheless, if either achieving fast model convergence is a top priority, or obtaining training speakers is challenging, then a viable compromise would be to train on a single WGN recording.

7.3 Simulation II - Number of Anchored Speakers

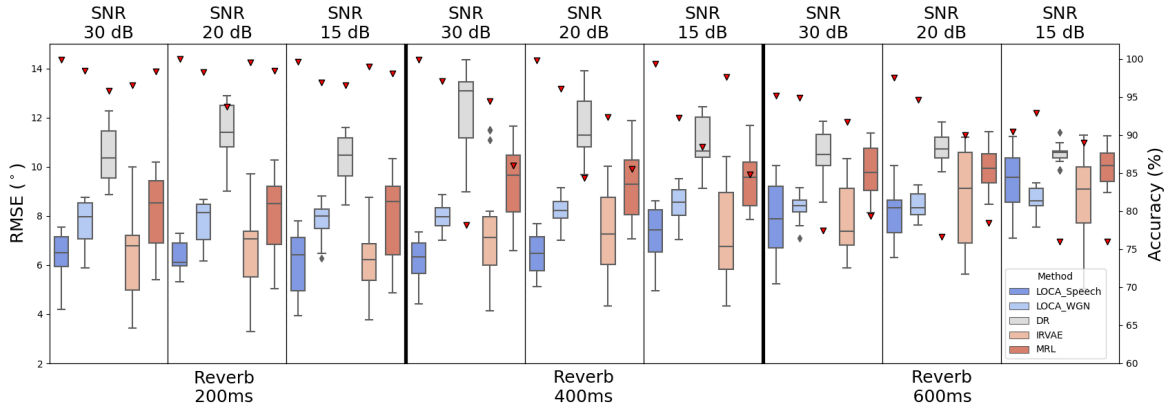
In the second simulation, we focus solely on LOCA's performance. In the previous simulation, only one speaker was used as an anchored speaker, providing a set of labeled anchor samples. We aim to investigate the extent to which this variable affects the results. In this scenario, we fix $N_L = 5$ labeled training samples per anchored speaker, equally spaced along



(a) 5 anchors



(b) 4 anchors



(c) 3 anchors

Figure 7.1: A comparison between the RMSE and accuracy measures for the different acoustic settings. Each sub-figure is attributed to a different amount of labeled anchor samples used. Within each sub-figure, a comparison is done for each possible combination of the examined SNR and reverberation levels. The boxes represent the distribution of RMSEs (Eq. 16) achieved for 20 different scenarios while the red triangles denote their tracking accuracy (Eq. 15).

the arc. Both the training and test data remain the same as in the previous simulation, with the only difference being the increase in number of anchored speakers.

Results and Discussion: The simulation results are depicted in Fig. 7.2. It is evident that while the improvement is less pronounced at low to mid reverberation levels, a significant enhancement is observed in highly reverberated acoustic environments. Specifically, notable improvement is observed when the number of anchored speakers is increased to three. This outcome effectively addresses the challenges we encountered earlier in the extreme acoustic conditions. While not requiring a significant amount of additional labels, we are now able to consistently achieve good results in these settings both in terms of tracking accuracy and RMSE.

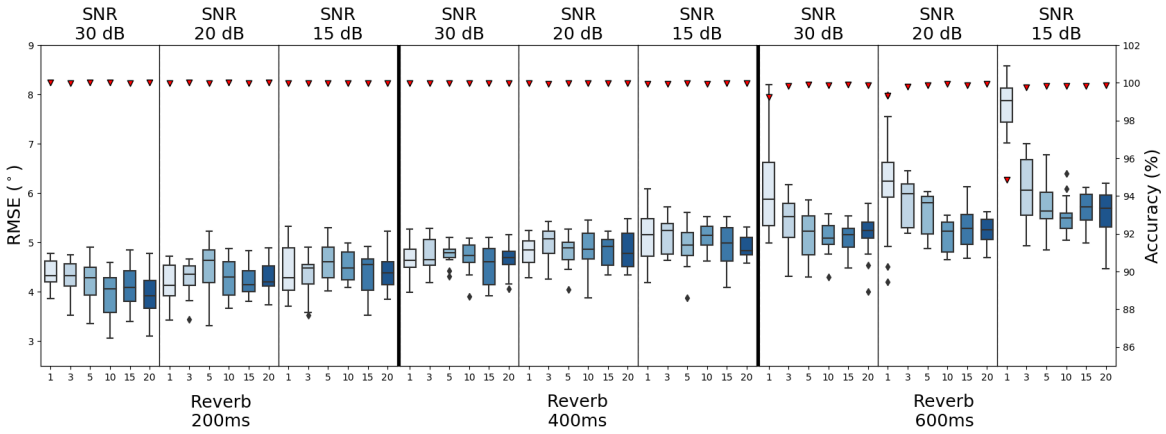


Figure 7.2: Performance comparison of the modified LOCA model that was trained on speech data (*LOCA-Speech*) when evaluated over various amounts of anchored speakers, demonstrated for the discussed reverberated and noisy conditions. For each anchored speaker, 5 uniformly spaced anchor samples are drawn from the arc. The boxes represent the distribution of RMSEs (Eq. 16) achieved for 20 different scenarios while the red triangles denote their tracking accuracy (Eq. 15).

7.4 Simulation III - Generalization to Test Context-Frames

When dealing with the task online dynamic speaker tracking, a crucial aspect to consider for the proposed solution has to be its ability to perform well for various levels of latency. Consequently, we found interest in exploring LOCA's ability to generalize its predictions for different quantities of context frames utilized during the testing phase compared to those used during the training phase. For this simulation, we implement the same training methodology used in Simulation I. Similar to Simulation I, only a single anchored speaker is providing anchors for training. The primary difference lies in the evaluation of LOCA's performance across different test datasets. These test datasets differ only in the amount of context frames used for iRTF estimation, where the considered values for N_c^{test} range from 6 to 20.

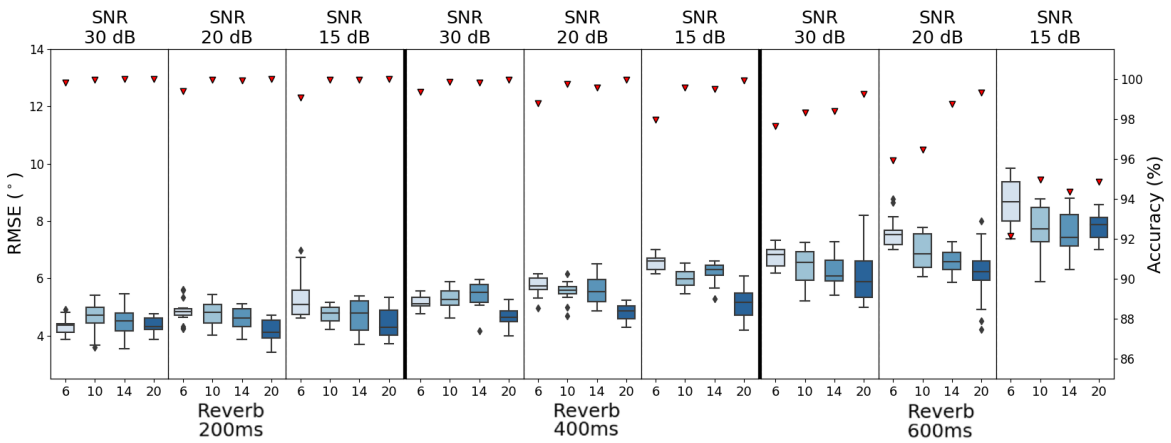


Figure 7.3: Performance comparison of the modified LOCA model that was trained on speech data (*LOCA-Speech*) when evaluated over different amounts of test context frames (N_c^{test}). Here, the amount of context frames used for training is $N_c^{train} = 10$.

Results and Discussion: Fig. 7.3 illustrates the evaluation results of LOCA across the various test datasets. In most acoustic settings, a slight improvement is noticeable in the cases where additional context frames are used to estimate the test iRTFs. This improvement is apparent both in accuracy and RMSE distributions. A slight decrease in performance is noticeable, however, in the case of $N_c^{test} = 6$, mainly in challenging reverberation and noise conditions. This decline is not necessarily attributed to data mismatch between training and test, since the mismatch is present in the higher N_c^{test} values as well. Rather, it may be due to the decreased accuracy in the iRTF acoustic feature estimation as the number of context frames decreases, especially in harsh environmental conditions. Nonetheless, it is evident that LOCA remarkably presents robust generalization capabilities, offerings significant flexibility in handling variations between the number of context frames during training and test phases.

7.5 Simulation IV Multi-Scenario Generalization

In our final simulation we aim to investigate LOCA's ability to perform a much more complex type of generalization. This generalization, which is referred to as multi-scenario generalization, is essentially a spatial interpolation within the enclosure. Given an acoustic setting, LOCA is trained on dynamic speakers drawn from a set of training scenarios. The task is then to perform tracking for speakers drawn from new scenarios in the enclosure, that were not seen during training. Recall that different scenarios are simply determined by the position and orientation of the microphone pair within the enclosure. We note that all of the scenarios for this simulation are constrained to the plane $z = 1.2m$. Furthermore, the scenarios under consideration involve speakers positioned at a radius of $r = 2$ m from the microphone pairs. The position of the interference noise source remains unaltered. The above idea is

conceptualized in Fig. 7.4.

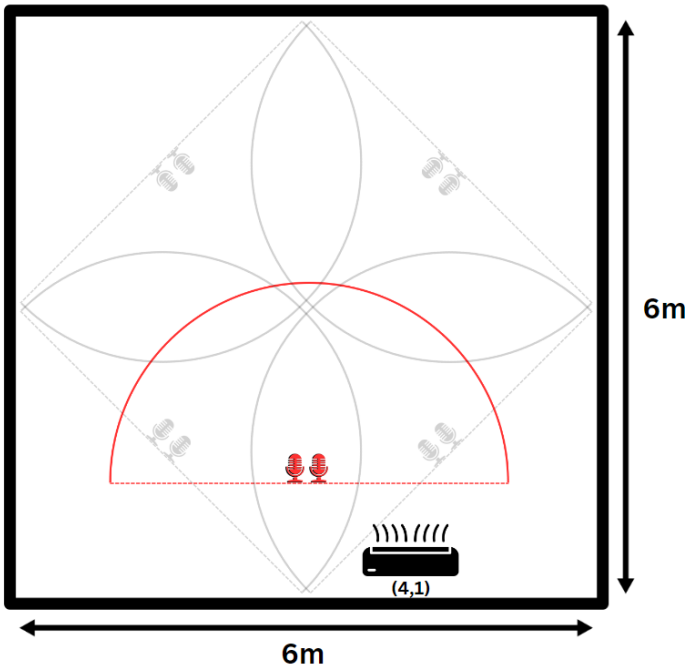


Figure 7.4: An abstract visualization for Simulation (7.5) - multi-scenario generalization. Our training data consists of speakers traveling along arcs of numerous scenarios (gray arcs). The test data belongs to scenarios that were not seen during training (red arcs). The interference source is located on $[x, y] = [4, 1]$, similar to the previous simulations. Both training and evaluation are conducted under the same acoustic setting (i.e., under the same reverberation and SNR levels).

7.5.1 Training Data

To create the training dataset, we considered 24 different training scenarios for any given acoustic setting. Looking forward to later successfully perform tracking in random scenarios, the picked scenarios were chosen such that the entire room was covered in the most symmetrical manner while considering the different possibilities for microphone orientations and positions. Each training scenario includes 7 random speakers, drawn from the TIMIT dataset. In a similar manner to Simulation I, we use the extended speaker recordings together with the signal generator package for MATLAB. The speakers were simulated with a constant velocity of 0.25 m/s, this time while conducting back-and-forth traversal along the arc. After applying Algorithm 1 for training-bursts collection with $N_c^{train} = 10$ train context frames, each training scenario provided approximately $9,100$ bursts, resulting in a total of $218,400$ bursts from the entire acoustic enclosure. For this simulation, which is already very challenging, we allow each of the 7 speakers to provide 5 equally-spaced anchor samples in each training scenario.

7.5.2 Test Data

For evaluation we consider 24 test scenarios that the model has not seen during training. In each scenario, 10 speakers are considered for evaluation. Once again, we are interested in evaluating the model's performance over multiple velocities, examining speaker velocities of $[0.25, 0.55, 0.85, 1.15]$ m/s. For each speaker, the iRTF features are estimate using $N_c^{train} = 20$, of which 20 samples are withdrawn per each of the four speaker velocities. This results in a total of $19,200$ test samples per acoustic setting.

7.5.3 Results and Discussion

In this simulation it is important to note that the training is conducted collectively using data extracted from multiple scenarios within the enclosure. This differs from our previous simulations where for each acoustic setting, models were trained and evaluated on the same scenarios individually. In the current setup, both the training and test datasets are mutual to the entire enclosure. Consequently, the results are not visualized through boxplots from a Monte Carlo simulation as done previously, but are rather represented in a tabular format, showcasing the performance measures achieved across the combined test dataset. The baseline model used for comparison is the IRVAE discussed earlier. The MRL approach was not considered for the task, as the great number of training samples is an obstacle for kernel based methods. Additionally, multiple attempts to utilize the DR model for the task approach were not successful at all.

Table 7.1 presents the simulation results. Aside from showing the results of the current experiment (Cur.), we include the nominal results (Nom.) in Experiment I to give a broader intuition of the difference in performances when dealing with the different tasks. It is worth noting that while IRVAE demonstrates some success in this challenging task, LOCA presents truly impressive results. As observed, both approaches generally experience a decline in performance as the noise and reverberation levels increase. Inspecting LOCA, we see that as expected, both the prediction accuracy and the prediction RMSE are negatively affected in this setup, in comparison to previous simulations. However, in almost each acoustic setting, LOCA achieves prediction accuracy of over 90%. A single acoustic setting which, sets the greatest obstacle for LOCA is the challenging setting of SNR $= 15$ dB and $RT_{60} = 600$ ms. Under these conditions, the model learns a poor and unreliable embedding for the manifold, and consequently, predictions significantly diverge from the true positions.

7.6 Ablation Study

As stressed earlier, in this manuscript, we have included several methodological adjustments to the LOCA training framework aimed at achieving more consistent convergence. In [59] a post-training calibration was suggested for incorporating the real-world coordinates into the learned embedding. Later, in [1], the single-step minimization of the loss function was

Model	SNR (dB)		200 (ms)	400 (ms)	600 (ms)
LOCA	30	Cur.	6.5 (100)	7.5 (97)	8.9 (93)
		Nom.	4.4 (100)	4.7 (100)	6.1 (99)
	20	Cur.	6.2 (100)	8.6 (95)	8.5 (94)
		Nom.	4.2 (100)	4.8 (100)	6.2 (99)
	15	Cur.	6.0 (100)	8.7 (90)	10.1 (64)
		Nom.	4.5 (100)	5.1 (100)	7.4 (98)
IRVAE	30	Cur.	7.1 (98)	9.2 (89)	9.2(81)
		Nom.	5.0 (99)	5.6 (98)	6.8 (96)
	20	Cur.	7.7 (97)	9.4 (90)	9.2 (79)
		Nom.	4.9 (99)	5.4 (97)	7.0 (95)
	15	Cur.	8.3 (92)	9.6 (86)	9.7 (77)
		Nom.	4.9 (98)	6.0 (98)	7.4 (9.6)

Table 7.1: Comparison between the modified LOCA and IRVAE in the multi-scenario generalization experiment. the results are represented for the current (Cur.) experiment in terms of median RMSE in cm (and accuracy percentage). The nominal (Nom.) results are also included, showcasing the methods' performance in Experiment I.

suggested, however, the post-training calibration approach was adopted as well. In the current work we also include the single-step minimization approach as an adjustment for the loss term, alternatively to the multi-step minimization approach. The *main* modification we introduce, however, involves incorporating the additional anchoring loss term into the optimization process.

In the following brief ablation study, we demonstrate the training stability of our proposed approach by comparing it with two ablated versions. The first ablated version, referred to as 'Multi-step minimization', is identical to our proposed approach except that it integrates multi-step minimization of the loss function. The second ablated approach, which is referred to as 'Post-training calibration', also retains the core components of our approach, but includes post-training calibration of the embedding instead of the proposed additional anchoring loss term. By comparing the performance of these ablated approaches with our proposed approach, we aim to assess the impact of each component on the overall effectiveness of the LOCA framework."

The models are trained and evaluated using our 1-dimensional data, following the performance measures discussed earlier. We conduct a Monte Carlo simulation for the task, mirroring the approach used in Simulation I and utilizing the same training and test datasets. Specifically, we consider an acoustic setting with reverberation and noise levels of $RT_{60} = 600$ ms and 30 dB respectively. The results are presented in Fig. 7.5. It is evident that the proposed approach achieves both the lowest median RMSE and the narrowest whiskers. In comparison, the ablated versions achieve slightly inferior results in terms of median RMSE, with significantly wider and skewed RMSE distributions observed across the 20 different scenar-

ios in the enclosure. These findings indicate that the proposed approach leads to a remarkably more stable convergence overall.

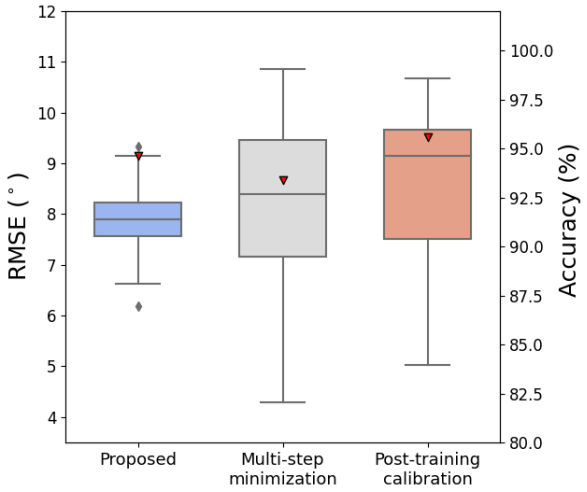


Figure 7.5: An ablation study on the stability of training for the different training approaches of LOCA. The results shown are attributed to the acoustic setting of $RT_{60} = 600$ ms and 30 dB SNR.

8 Conclusions

8.1 Summary

In this thesis, we harness recent advances in DNN-based manifold learning to address real-life problems in acoustic reverberant environments. The work encompasses two main applications: acoustic scene mapping and semi-supervised dynamic speaker tracking.

For the acoustic scene mapping task, we demonstrate the applicability of the RTF as a proper feature vector, encapsulating relevant spatial information. By leveraging the natural structure of the data and local relations between samples, we avoid the severe performance degradation typical of other localization approaches. Our simulation results show the ability to extrapolate to regions not seen during training, significantly reducing run-time requirements compared to existing schemes. Additionally, we confirm the robustness of the approach even under mild to high reverberation levels.

Dealing with the speaker tracking task, we introduce two novel adjustments to the original LOCA algorithm to further optimize the training process, namely the anchoring loss term and the single-step minimization. By utilizing high-dimensional instantaneous-RTF acoustic features lying on a manifold, the model learns an embedding that is isometric to the latent manifold underlying the data. Our approach aims at utilizing a minimal amount of labeled data, a resource which is often scarce. To achieve that, we apply the synthetic aperture training paradigm to exploit local relations between subsequent samples, effectively leveraging the movement of the speakers during training phase. A series of simulations showcases the ability of the proposed method to track speakers under challenging conditions of noise and reverberation, as well as the influence of the amount of labeled samples. Furthermore, the demonstrates remarkable generalization capabilities, enabling predictions not only for higher speaker velocities, but also for varying levels of latency, and even for different configurations within the acoustic enclosure.

8.2 Future Directions

Collaborative Autoencoder Networks for Multi-Source Acoustic Scene Mapping

Dealing with the problem of acoustic scene mapping, we used sampled data from two sound sources to map the enclosure, as it proved way more efficient when dealing with a wide-scale enclosure. Our way of merging information was simply to concatenate the input features, but it might be counterproductive when the dimensionality of the features is high, i.e., when N_{FFT} is large. However, one could think of a more clever way to merge the multi-source information. A possible idea would be to use a different autoencoder network for each one of the sound sources. The two networks could then be coupled with a shared latent embedding layer at the output of their encoders to learn coordinates

that would match each manifold from each source individually. The coupling could be done by adding an RMSE loss term between embeddings of the shared samples. Another idea, would be to treat it as a multi-modal problem where we try to learn two different embeddings such that the correlation between them is maximized. This could be done by implementing a Canonical Correlation Analysis (CCA) loss on the output of the encoders.

Integration with Multi-Modal Data

Integrating multi-modal data from sources such as visual cameras, infrared, or LIDAR sensors with acoustic data can significantly enhance both the capabilities of acoustic scene mapping and dynamic speaker tracking. This approach leverages the strengths of different data modalities to provide a more robust and comprehensive understanding of the environment.

As discussed in Sec. 4.1 our proposed approach for acoustic scene mapping reduces performance degradation in reverberated settings. However, degradation in performance is still visible, especially in the presence of high reverberation and in complex environments where physical obstacles block and distort the acoustic signals. By integrating visual data, the system can gain additional spatial information that complements the acoustic data. Visual data can help identify physical boundaries that are imperceptible through sound alone. One could develop a multi-modal fusion algorithm that combines the strengths of each modality. We can construct a neural network designed to take both RTF acoustic features and visual features as input, learning a joint representation that captures the spatial relationships in the environment more accurately.

A similar approach could apply to dynamic speaker tracking, where using noisy and reverberant speech alone still presents significant challenges. One could design a multi-modal tracking neural network based on LOCA, where visual sensors provide initial localization and movement prediction, while LOCA uses acoustic data to refine the tracking by providing detailed sound source localization. A deep learning model could be trained to fuse these inputs, learning to weigh the contributions of each modality dynamically based on the reliability of the data in different acoustic and visual scenarios.

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תקציר

מטלפונים חכמים ורמקולים חכמים ועד למכשירי שמיעה ומערכות מעקב, מכשירים המצוידים במיקרופונים נמצאים כיום בכל עבר. כתוצאה מכך, ישנו ביקוש הולך וגובר לשיטות מתקדמות של עיבוד אותות שמע במטרה לפרש בצורה נכונה מידע אקוסטי בסביבות שונות. אחד מהאתגרים המשמעותיים ביותר בתחום הוא הדהוד, אשר עשוי לעוות אותות שמע באופן חמור ולפגום בביצועים של שיטות עיבוד אותות מסורתיות. מיפוי של זירה אקוסטית ואיכון דובר נע בסביבות מהדהדות הן משימות קריטיות ליישומים רבים, החל ממציאות רבודה ועד רובוטיקה וטכנולוגיות סיוע.

החלק הראשון של עבודה זו מתמקד בשחזור זירה אקוסטית, במטרה למפות סביבה תוך שימוש אך ורק באותות האקוסטיים הנמדדים על ידי צופה. שיטות קלאסיות לשחזור זירה אקוסטית ואיכון דובר נע לרוב מתבססות על הערכת הפרש זמן הגעת האותות, גודל שבדרך כלל רגיש מאוד להדהוד ולרעש מדידה. כאשר הסביבה האקוסטית סטטית, אנו טוענים כי שינויים בקלט האקוסטי נובעים בעיקר משינויים במיקום המקלט, מה שמרמז כי הקלטים האקוסטיים אשר נמדדים ומיוצגים במימד גבוה למעשה שוכנים על גבי יריעה במימד נמוך. תוך ניצול רעיון זה, אנו מיישמים כלי חדשני ללמידת יריעה באופן בלתי מפוקח אשר מבוסס על תצפיות מרובות. אנו מציעים להשתמש ב- Local Conformal Autoencoder (LOCA) , אלגוריתם מבוסס רשת נוירונים עמוקה ללמידה של יריעה, כדי לנצל את המבנה הטבעי של המדידות האקוסטיות. הגישה הבלתי מפוקחת שלנו משתמשת במערך מיקרופונים נייד כדי לקלוט את הפלט של מקורות קול כפי שנמדד במיקומים שונים בתוך הסביבה אקוסטית. אנו מדגימים כי השיטה שלנו לומדת ייצוג איזומטרי של מיקומי המיקרופונים, המאפשר אקסטרפולציה לאזורים חדשים בהם עוד לא ביקרנו. אנו מאמתים את עמידות הגישה באמצעות סדרת סימולציות, ומוכיחים את עמידותה להדהוד בהשוואה לשיטות קלאסיות ולשיטות אחרות של הורדת מימד.

החלק השני של עבודת תזה זו מתמקד באיכון דובר נע, ובמסגרתו אנו שואפים לעקוב אחר כיוון ההגעה של סיגנל הדובר בזמן אמת בסביבות רועשות ומהדהדות. אנו מציגים גרסה חצי-מפוקחת של מודל LOCA יחד עם פרוצדורת אימון חדשנית של מנגנון מפתח סינתטי. באמצעות שיטה זו, אנו מנצלים את היתרונות של LOCA תוך ניצול האופי הדינמי של הבעיה. סדרת סימולציות נרחבת מאשרת את יעילות השיטה שלנו בסביבות דינמיות עם רמות שונות של רעש והדהוד, תוך הדגמת יכולת ההכללה שלה לתרחישים חדשים.

עבודה זו נעשתה בהדרכתם של פרופסור שרון גנות ודוקטור אופיר לינדנבאום
מהפקולטה להנדסה של אוניברסיטת בר אילן

אוניברסיטת בר אילן

רשתות אוטו-אנקודר עמוקות קונפורמיות-מקומית עבור מיפוי סביבה אקוסטית ואיכון דובר נע באופן מפוקח למחצה

עידן כהן

עבודה זו מוגשת כחלק מהדרישות לשם קבלת תואר
מוסמך בפקולטה להנדסה של אוניברסיטת בר אילן

רמת גן תשפ״ה